Manufacture and testing of jet engines:case study of AL31FP engine

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3 authors, including:

Vamsikrishna Undavalli
University of Alabama

Balakrishna Bodramoni
Amity University

AMITY INSTITUTE OF AEROSPACE ENGINEERING

STUDY OF AL31FP ENGINE MANUFACTURING ASSEMBLY & TESTING

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HAL KORAPUT

Undavalli.Vamsikrishna - 05

Bodramoni.Balakrishna - 24

5 th semester AIAE

AMITY UNIVERSITY

ACKNOWLEDGEMENT LETTER

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Now at last but not least we are thankful for entire HAL Koraput division employees for their patience in explaining their work in detail which helped us a lot to acquire great knowledge .We are not able to mention all the names of employees but we have to mention two employees **Shri Rajiv mohanthy** ,manager of dismantling bay and **Shri Suresh** inspection officer of Assembly section view room due to special care taken by them in helping throughout the project.

Yours faithfully

Undavalli Vamsikrishna

Bodramoni Balakrishna

AMITY UNIVERSITY

CERTIFICATE OF FACULTY GUIDE

This is to certify that **Mr.Undavalli Vamsikrishna** and **Mr.Bodramoni Balakrishna** students of 5^th semester B.TECH Aerospace Engineering, Amity University, Noida has carried out their internship on **STUDY OF AL31FP ENGINE MANUFACTURING , ASSEMBLY & TESTING** at Hindustan Aeronautics limited, Koraput (S.E.D) under my guidance and their work is satisfactory.

Mr. SAQUIB REZA

AIAE department,

Amity University, Noida

JOB WORK

VAMSIKRISHNA	BALAKRISHNA
• History and Invention of Jet engine • Major components of Jet engine • Introduction to AL31FP and study of its major components • Manufacturing process involving welding section,Heat treatment,Electroplating. • Assembly section with balancing section and dismantling. • Test house ➔ Testing parameters • Comparative study of HAL Koraput engines. • Personal interaction with Airforce personals,CRI,CRE • Research and Development • Central laboratory ➔ X-ray diffraction ➔ NDT	• Introduction to HAL industry Koraput division • Major components &aggregates of AL31FP with their functioning • Manufacturing process involving investment casting, forge, sheet metal shops. • Assembly section with aircraft gear box,engine gear box,turbostarter. • Test house ➔ Test bed installation and testing procedure . • Central Laboratory ➔ Metallurgy ➔ Mechanical ➔ Pyrometry ➔ Fuel and oil ➔ Spectroscopy anyalsis

INTRODUCTION TO HAL AND TO PROJECT

Hindustan Aeronautics Limited is the one of the Asia premier organization of present Aerospace industry.

HAL Koraput division is manufacturing and overhauling unit for various aero engines.

As per the present scenario the division manufactures only AL31FP engine for SUKHOI- 30MKI aircraft.

It overhauls

AL31FP for SUKHOI -30 MKI

RD33 for MIG -29

R29B for MIG -27M

R25 for MIG -21BISON

All the above stated engines comes under turbojet engine category which are a type of jet engine.

Our project work is carried here for 6 weeks covering all the sections from research and development section to final dispatch section in order to acquire industrial oriented procedure.

This project helps the people to understand from the basics of Jet engine because we initially covered a section JET ENGIENS in order to give the knowledge of JET ENGINE ,its components and its working.

Then next phase of our work is shifted to industry different shops involved in manufacturing and overhauling of JET ENGINE including testing section.It is briefly mentioned all the process involved in industry step wise in order to understand easily.

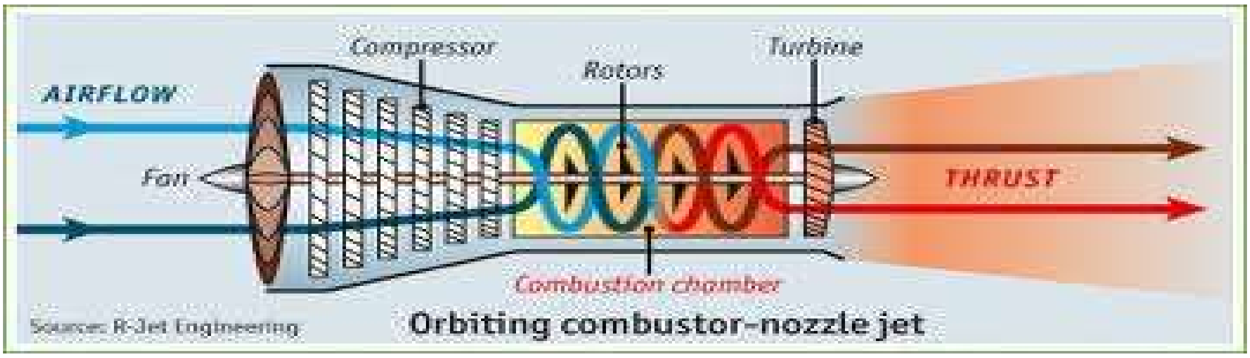
Finally phase we covered our work with central laboratory where all the physical and chemical analysis is done and interaction with AIR FORCE engineers,R&D engineers,CRI,CRE.

JET ENGINES

JET "means rapid stream of liquid or gas forced out of a small opening".

DEFINITION OF JET ENGINE:

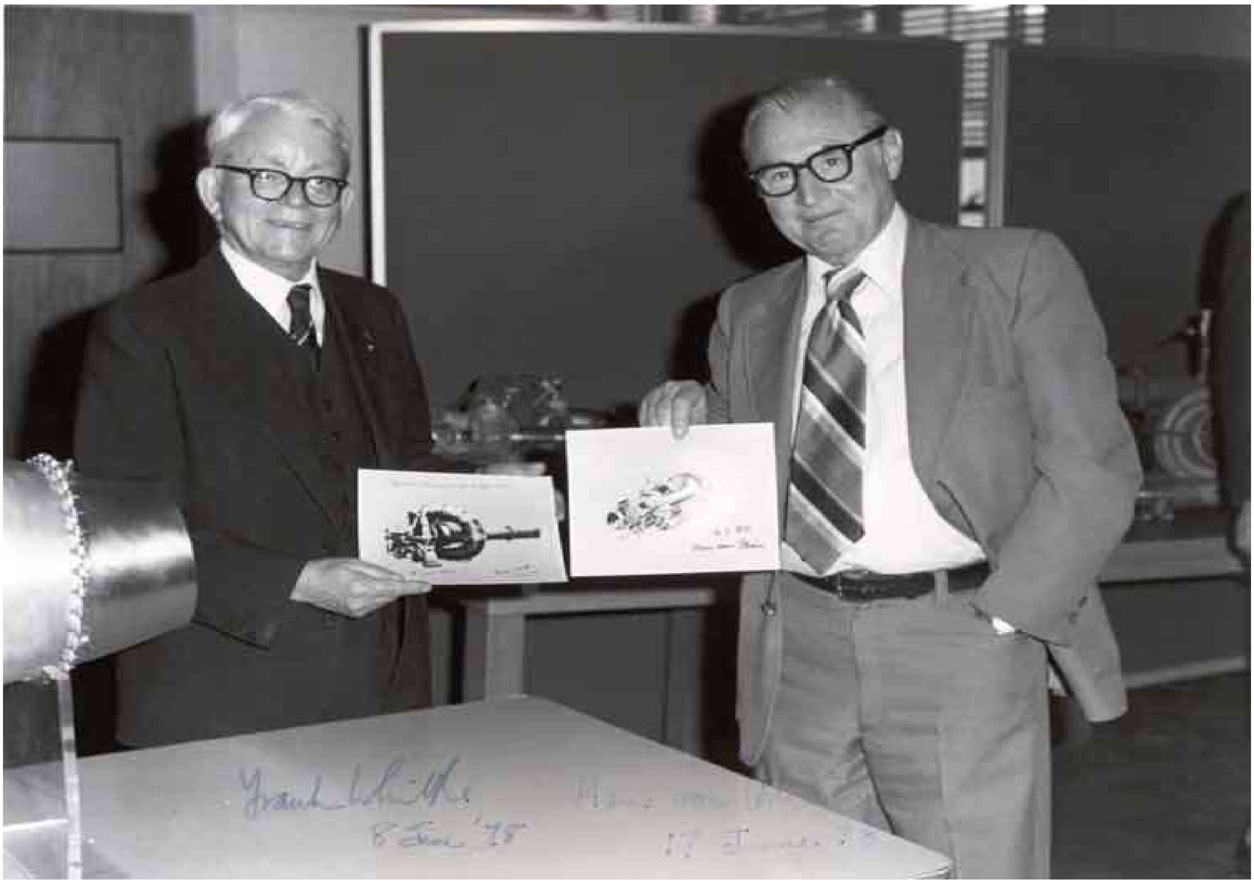
"JET ENGINE is a reaction engine that discharges a fast moving JET through small cross sectional area (nozzle) by jet propulsion which creates thrust in accordance to Newton's laws of motion".



This broad definition includes a turbojets, turbofans, rockets, ramjets and pulsejets etc.

HISTORY AND INVENTION: Dr. Hans von Ohain and Sir Frank Whittle are both recognized as being the co-inventors of the jet engine. Each worked separately and knew nothing of the other's work.

A young Royal Air Force man, Sir Frank Whittle, had presented to the Air Ministry a design for a jet engine in October 1929. They were unimpressed and rejected his idea. Regardless of this setback, Whittle still patented his "turbojet engine" he developed his ideas further and on 16 January 1930 in England, Whittle submitted his first patent (granted in 1932) and first to register patent. In 1936, he set up a company called Power Jets Ltd. In 1937, using newly available alloys that were strong and light, he produced the first viable jet engine to be successfully tested in a laboratory and first flew in 1941. Dr.Hans von Ohain, a German scientist was granted a patent for his turbojet engine in 1936 and flew his Jet aircraft in 1939.



Sir Frank Whittle (left) and Dr.Hans von Ohain (right)

All jet engines works on BRAYTON thermodynamic cycle developed by George Brayton but, patented by Englishman John Baber in 1791.It is also sometimes known as the **Joule cycle**. It is a constant pressure cycle accomplishing the same operations as in Otto cycle, but all operations are continuously with uninterrupted flow of power.

Ideal Brayton cycle:

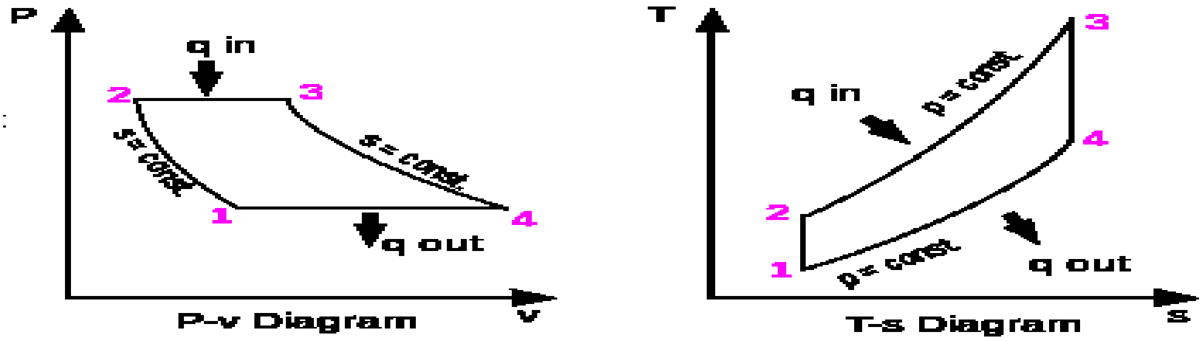
Processes of ideal Brayton cycle:

1-2 isentropic process - ambient air is drawn into the compressor, where it is pressurized.

2-3 Isobaric process - the compressed air then passes through a combustion chamber, where fuel is burned, heating that air—a constant-pressure process, since the chamber is open to flow in and out.

3-4 isentropic process - the heated, pressurized air then gives up its energy, expanding through a turbine (or series of turbines). Some of the work extracted by the turbine is used to drive the compressor.

4-1 Isobaric process - heat rejection (in the atmosphere).

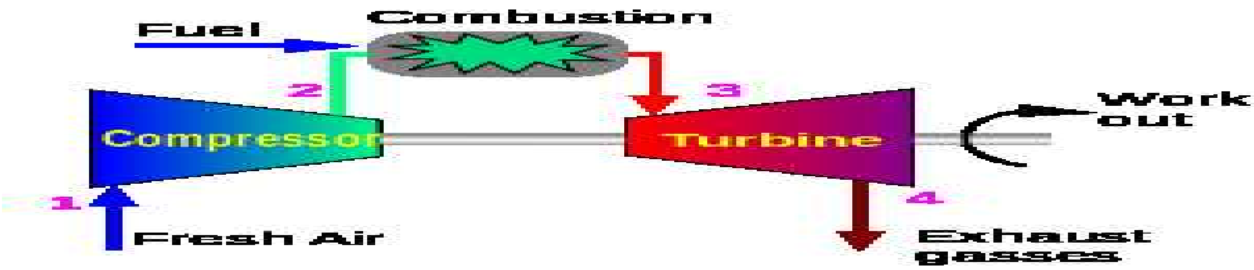


Pressure - Volume and Temperature – entropy diagrams

COMPONENTS OF JET ENGINE

Jet engine essentially consist of three elements in effect of conversion of heat energy to mechanical energy.

1. Compressor
2. Combustion chamber
3. Turbine



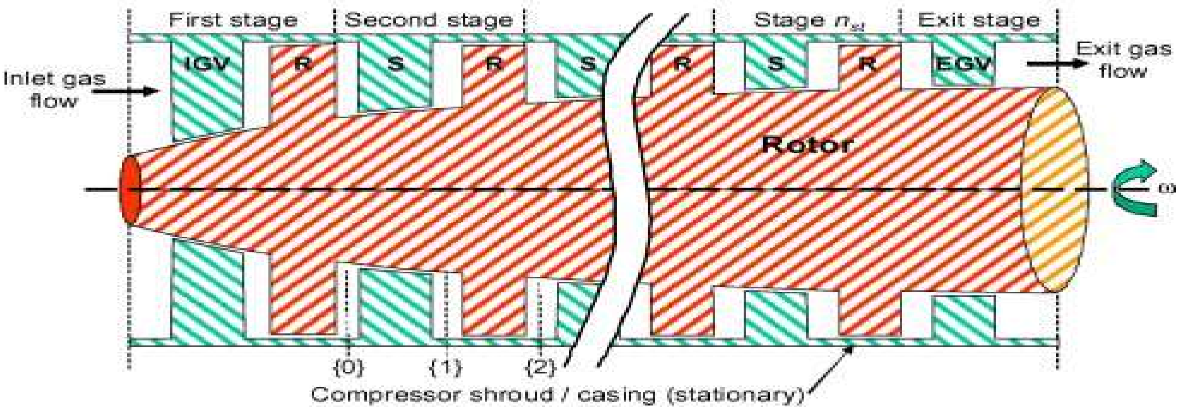
The Compressor is a rotating element which sucks in air at ambient pressure and temperature and compresses it to high pressure. The pressurized air enters into combustor, here air mixes with injected fuel. As the burning fuel releases heat temperature rises substantially. The high energy products of combustion expand in the turbine and their pressure and temperature drop as they do work in the turbine.

Compressor

There are two types of compressors –Axial flow compressor

-Centrifugal compressor

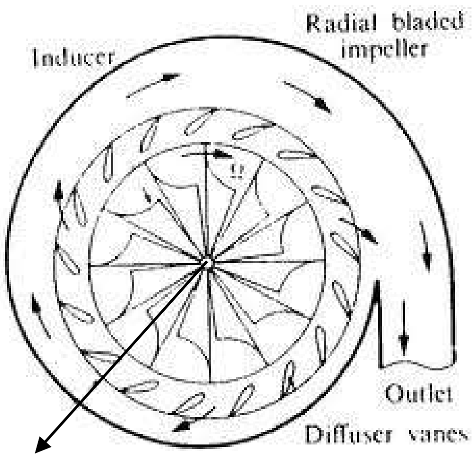
Axial flow compressor:



This type of compressor consists of series of rotating blades(R) and a row of stationary blades(S).Air speeds up in rotating blades and slows down in the fixed blades. The stationary blades act as diffusers and rotating blades increase the pressure.

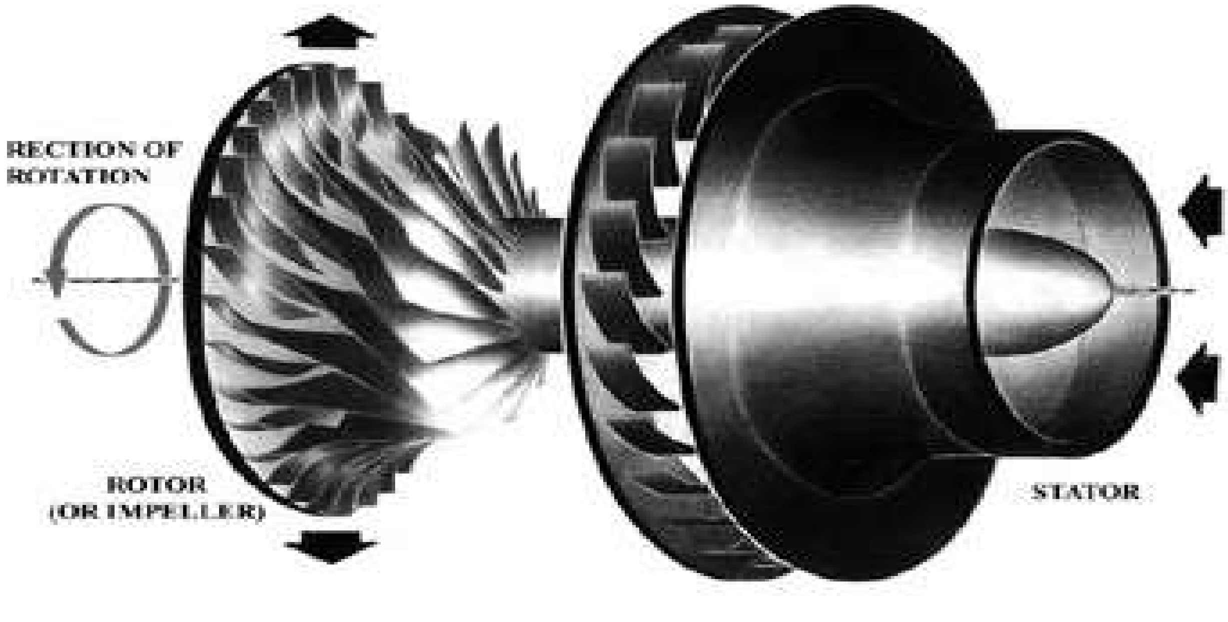
A row of fixed and a row of moving blades together are said to a stage and in each stage pressure increases 1.2 times initial stage pressure. The compressor blades must be accurately shaped, clean to achieve high compression efficiencies 95% range. As pressure rises specific volume of air decreases and so the height of blades decreases from stage to stage. A multistage compressor is capable of achieving a pressure ratio of 1:25 in large units.

Centrifugal compressor:



Inlet of air (center)

Air enters at the center of rotating compressor. The centrifugal force acting on the impeller forces air to move radically into the stationary vanes which act as diffusers. The air then passes through a collector which also act as diffuser, which pressurizes by slowing the flow of air



CENTRIFUGAL COMPRESSOR STAGE

Combustion Chamber

Combustion chamber burns fuel with pressurized air delivered by the compressor. Only 25-30% of this air actually contributes to fuel oxidation and the remaining air is known as secondary air that is used to cool the gas going to the turbine .The centigrade. There are three basic types of combustion chambers are used in Gas turbines

1) Can-type OR Multiple combustion
2) Annular type
3) Can-Annular type OR Tubo- annular combustion

Temperature after combustion may rise to 1900 degree

Can type or multiple combustion chamber

It is made up of individual combustion chambers. The air from compressor enters each individual combustor through an adoptor.Each combustion chamber is composed of two cylindrical tubes, the combustion chamber liner and outer liner,

Combustion takes within the liner. Air supply is controlled Louvers

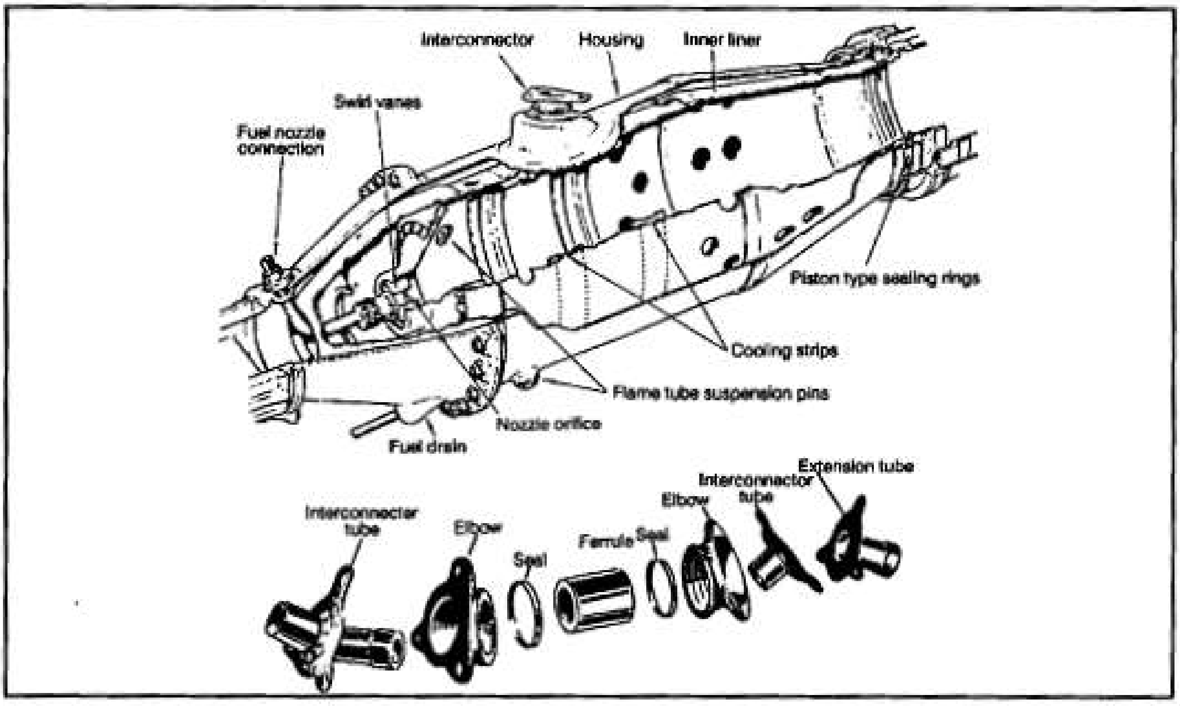
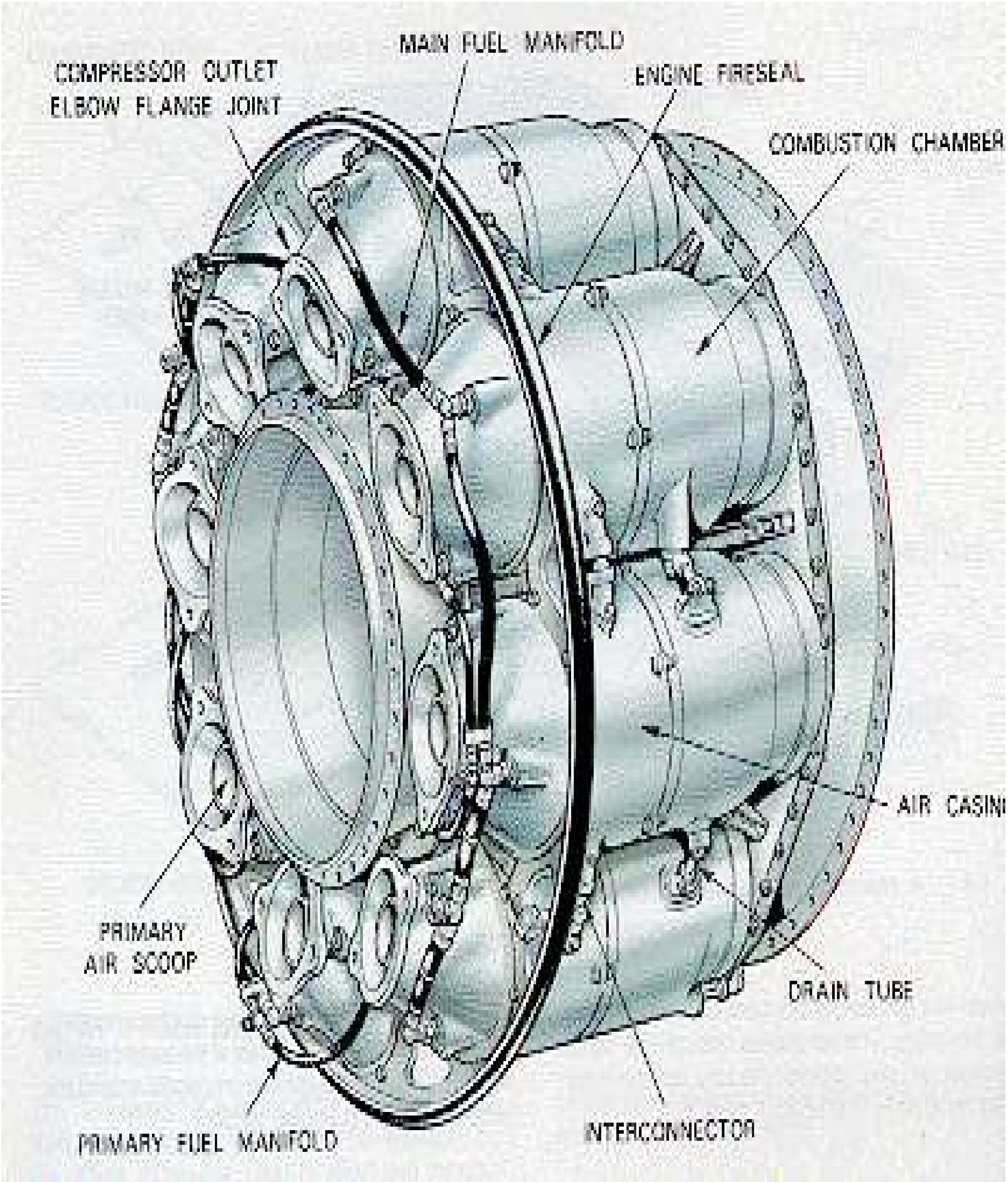
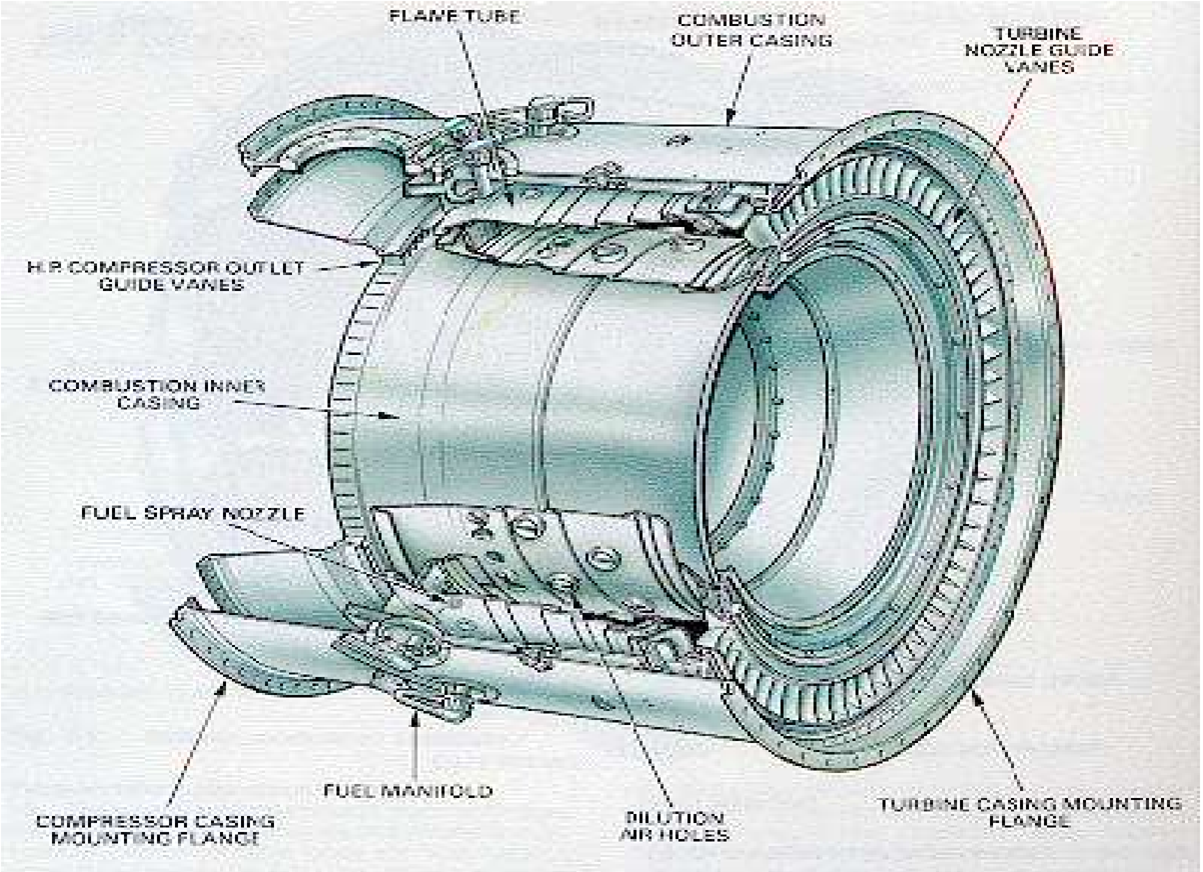


Figure 3-12. Can-Type Combustion Chamber Components



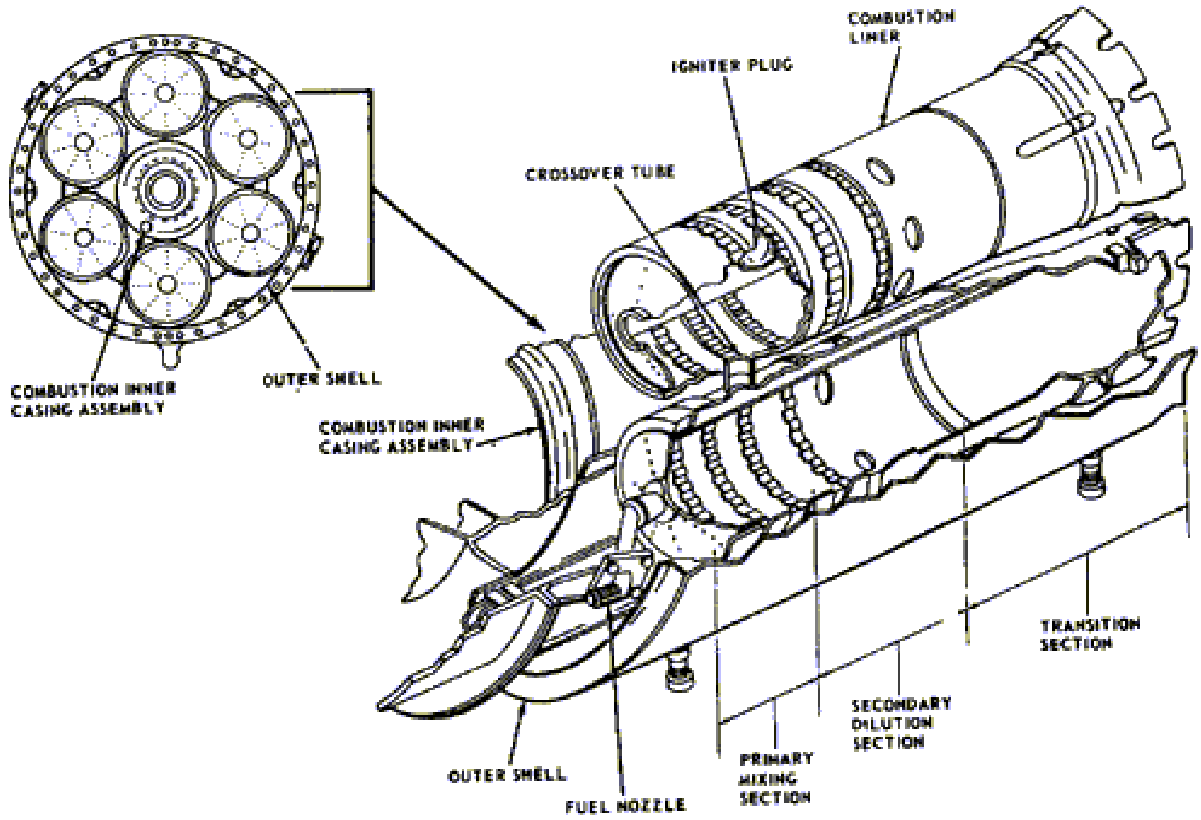
Annular type combustion chamber

An annular is mounted concentrically inside an annular casing. It results in compact design and lower pressure loss than other designs. Enough space is left between outer wall and combustion chamber housing to permit the flow of cooling air from the compressor

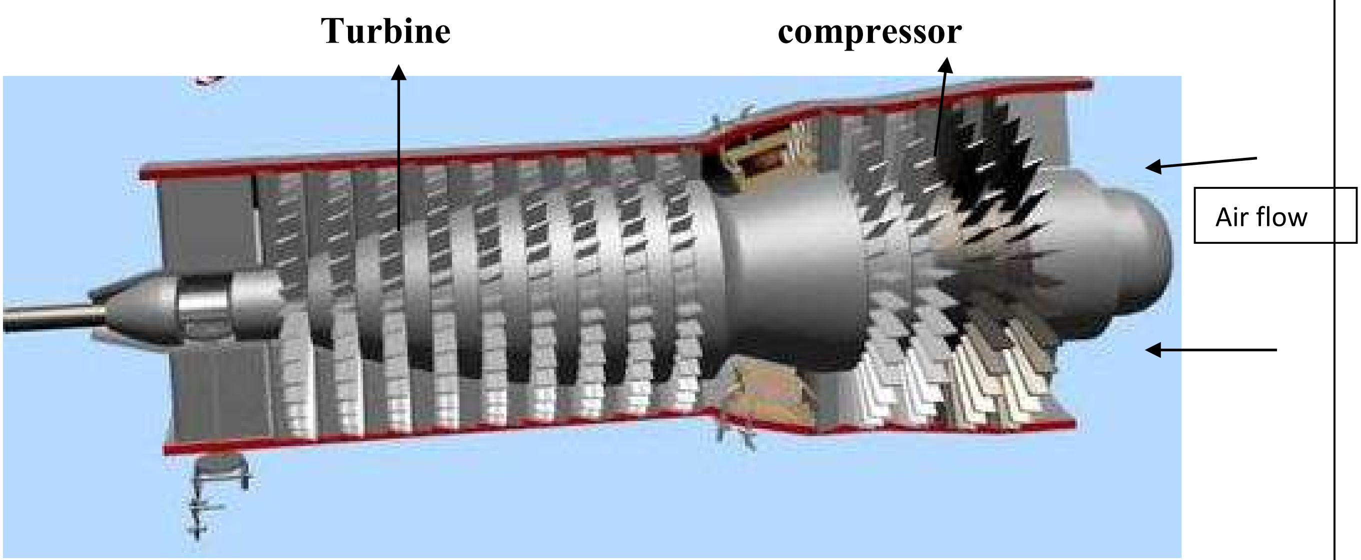


Can annular type

The combination of Can and Annular is Can-Annular combustion chamber. It consists of an outer shell with number of individual cylindrical liners mounted about the engine axis. The combustion chambers are completely surrounded by airflow that enters the liners through louvers. The flame is carried by through cross tube to other liners.



Turbine



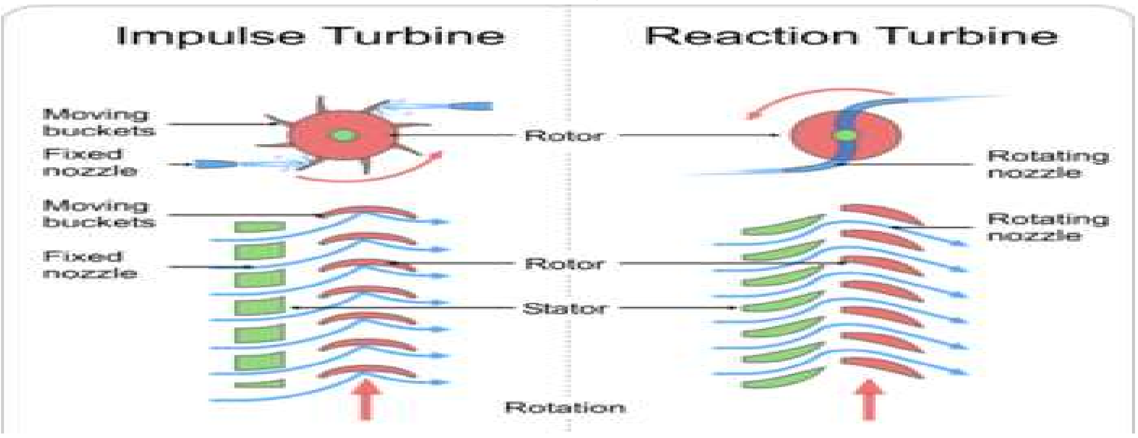
The hot gases pass through turbine after combustion chamber to produce power i.e. To rotate the shaft/compressor/propeller est. Turbine is mainly made up blades which play a crucial role in power generating.

1) Impulse

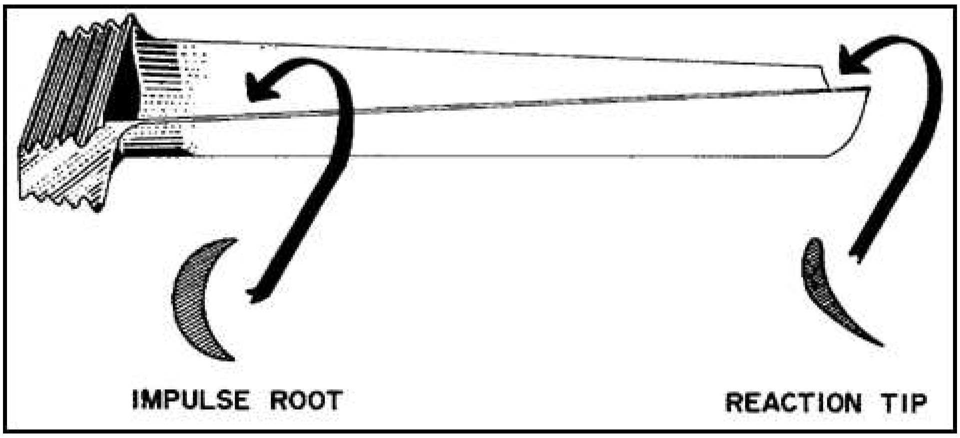
2) Reaction

In Impulse turbine stage the stationary vanes are called nozzles and moving are called buckets ,where all the enthalpy drop of this stage takes place creating a high velocity which impinges on the moving blades creating an impact.

In reaction type the total enthalpy drop is divided between fixed and moving in different proportions which shown in fig.



In gas turbines the gases expand as they pass from one stage to another stage, the temperature and pressure drops and specific volume of gas increases so the blade size must be increased to accumulate large quantity of gas .So as a result blade profile is changed by twisting the blade from root to tip.



Mounting of moving turbine blades

The blades are attached to turbine wheel by using ‘fir tree “construction at the root in open tip and shrouded tip .The blades are kept from moving axially either by rivets a special locking device. The shrouded blades have more aerodynamic advantage as shrouding helps to prevent hot gases over the tip of the blade and excessive blade vibrations.

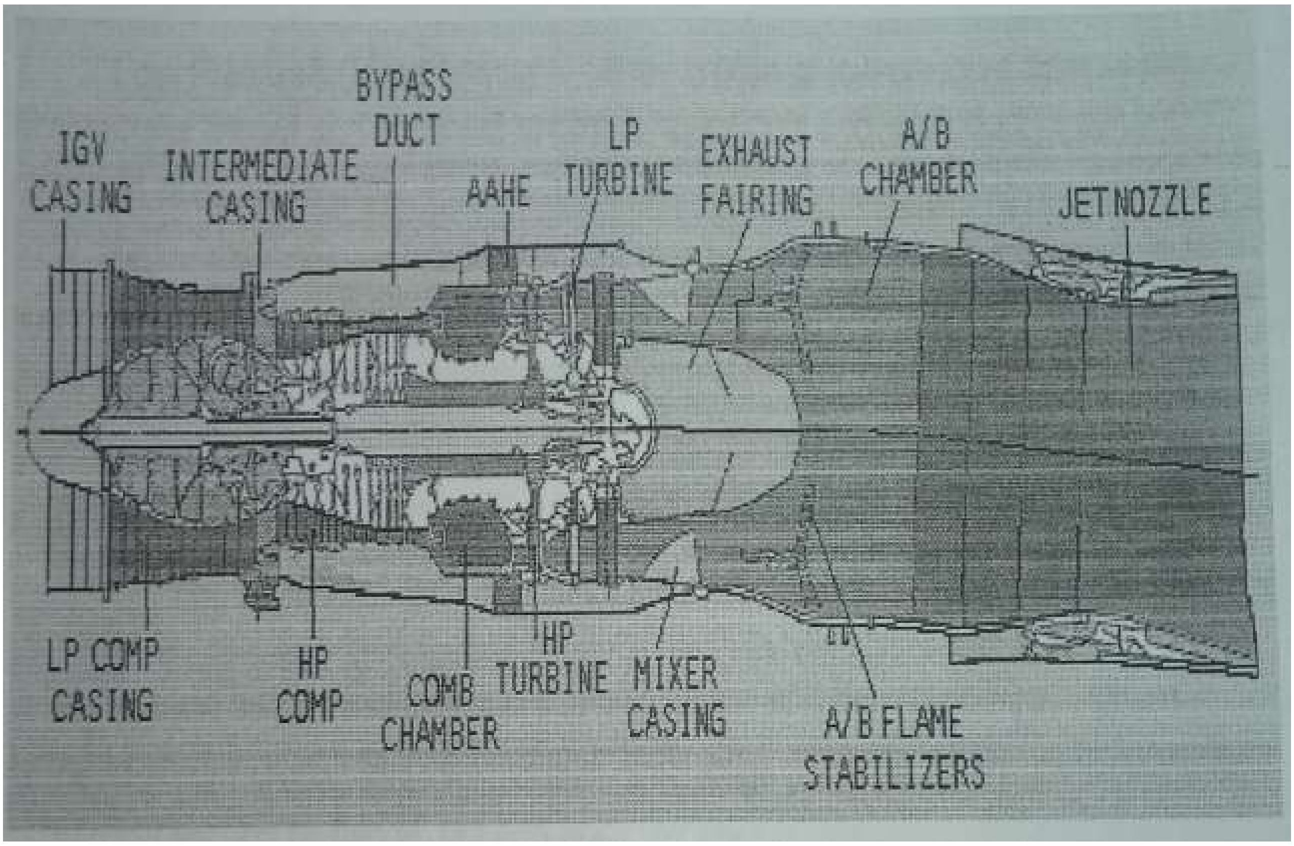
AL31FP

AL 31 FP engine is designed and developed by Lyulka, of Russian based company and now HAL is manufacturing the engines under license.

AL 31 FP engine, stands for

A- Name of designer- Arkhip Mikhailovich Lyulka
L- Name of company-Lyulka, now NPO Saturn
31-stands for series
F-stands for after burner (Farsa in Russian language)
P-stands for thrust vectoring nozzle (Povorothoye in Russian language)

The AL 31 FP is a turbojet engine with having a special features of Variable LPCR guide vanes, by-pass duct from LPCR and variable HPCR stator blades of Stages 1, 2, 3 and thrust vector nozzle by tilting angle of 14 degree.



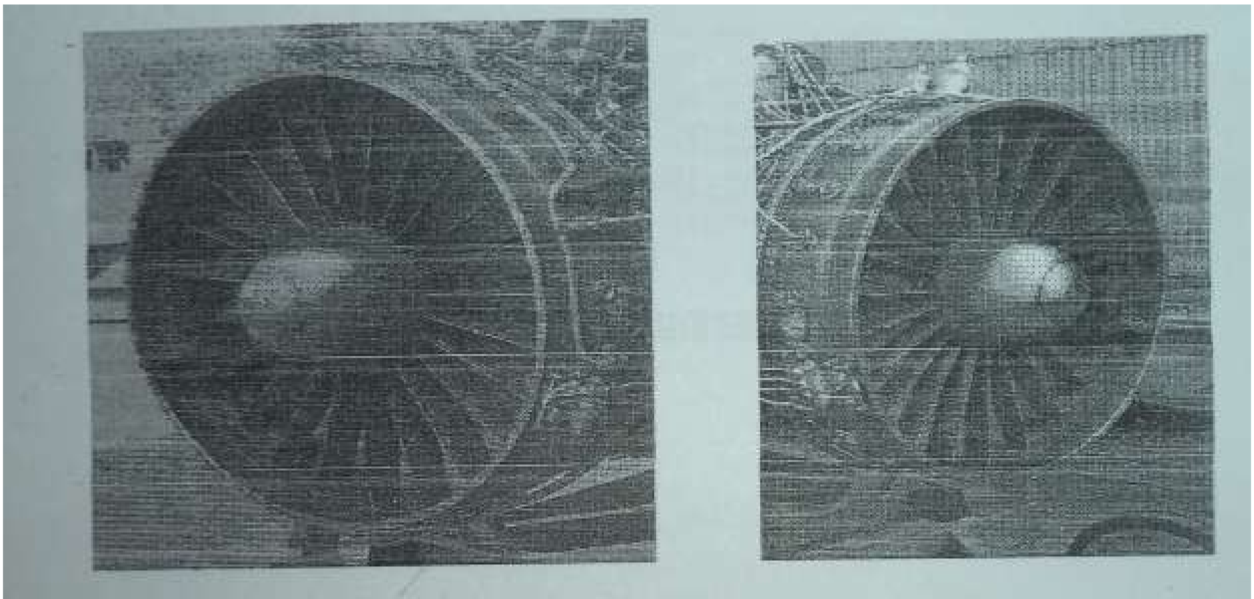
PICTURE OF AL31FP Sectional view

INLET GUIDE VANES

In AL31FP the inlet guide vanes are of variable so that which there local angle of incidence can be changed according to velocity and altitude of flight automatically by fly by wire control unit. There is a temperature sensor at this section as this is the intake section. If the temperature is very low that there is ice formation it activates anti icing system, which blows the hot air tapped from turbine section into the hollow section of guide vanes and heats up the intake air.

The guide vanes are manufactured by titanium based alloy by forging and relevant machine process as per technology. The outer casing is made up of aluminum alloy by forging and machine process.

- In RD33, R25U, R29B the inlet guide vanes are not of variable type.



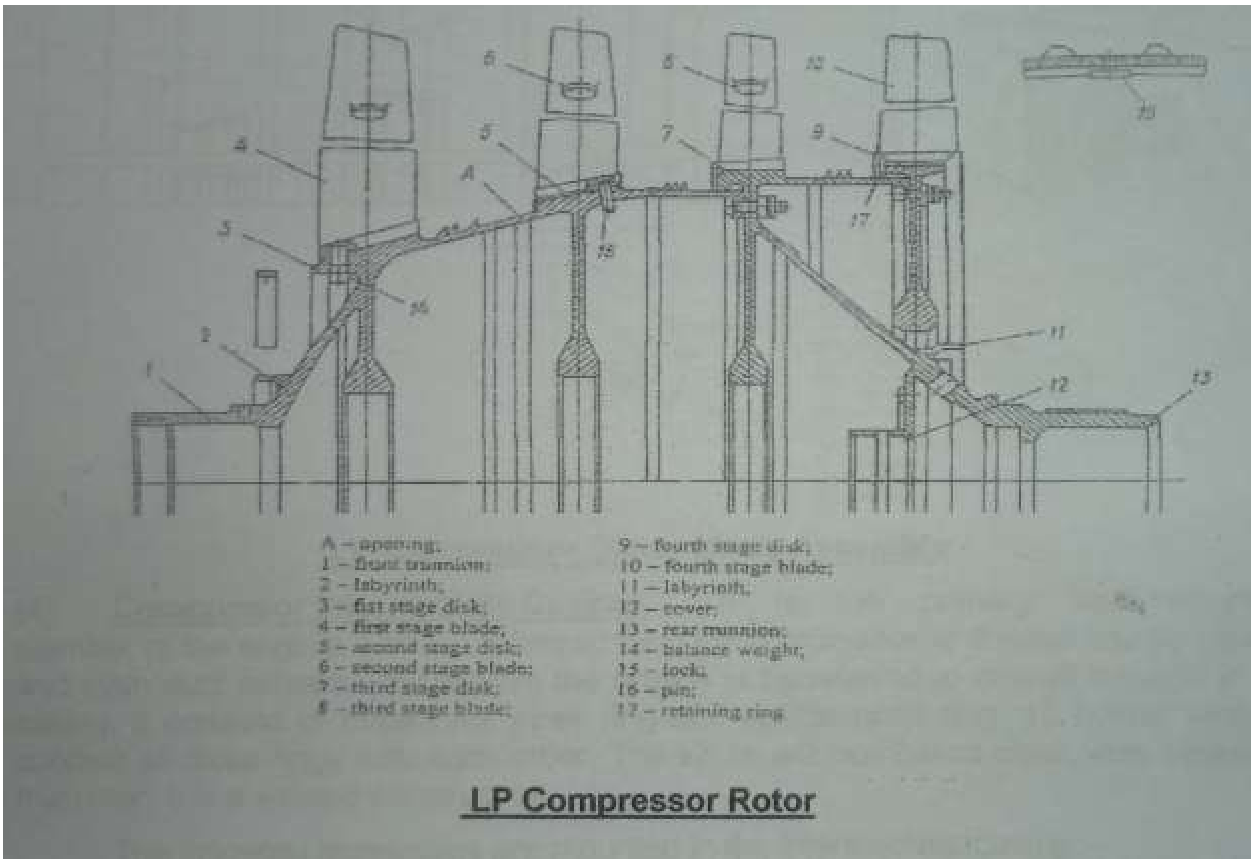
Inlet Guide Vane Assembly

COMPRESSOR SECTION

In AL31FP the compressor is 13 stage axial compressor out of 13 stages 4 stages are low pressure compressor and 9 stages are high pressure compressor. Compressor ratio is 23:1

- In RD 33 the compressor is 13 stage axial compressor with 4 stage LPCR and 9 stage HPCR.Compression ratio is 20:1.
- In R29 B the compressor is 11 stage axial compressor with 5 stage LPCR and 6 stage HPCR.Compression ratio is 12:1
- In R25U the compressor is 8 stage axial compressor with 3 stage LPCR and 5 stage HPCR.Compression ratio is 9.5:1

Low pressure compressor:



LP Compressor Rotor

Low pressure compressor is 4 stage axial one with a compression ratio of 3.5:1 .Each stage of compressor is having a series of rotor and stator blades. Rotor blades are used to compress the air by transferring the kinetic energy to air molecules and stator is used guide the air to the next stage by stopping the air which results in pressure rise.

Low pressure compressor is connected with low pressure shaft to low pressure turbine and this LPT drives the LPC.This interconnection of LPC, LPT with a shaft is low pressure spool LPS.

LPC is generally made of blades which pressurize the air and discs which are used to mount the rotating blades and an outer casings which is used to mount the stator blades

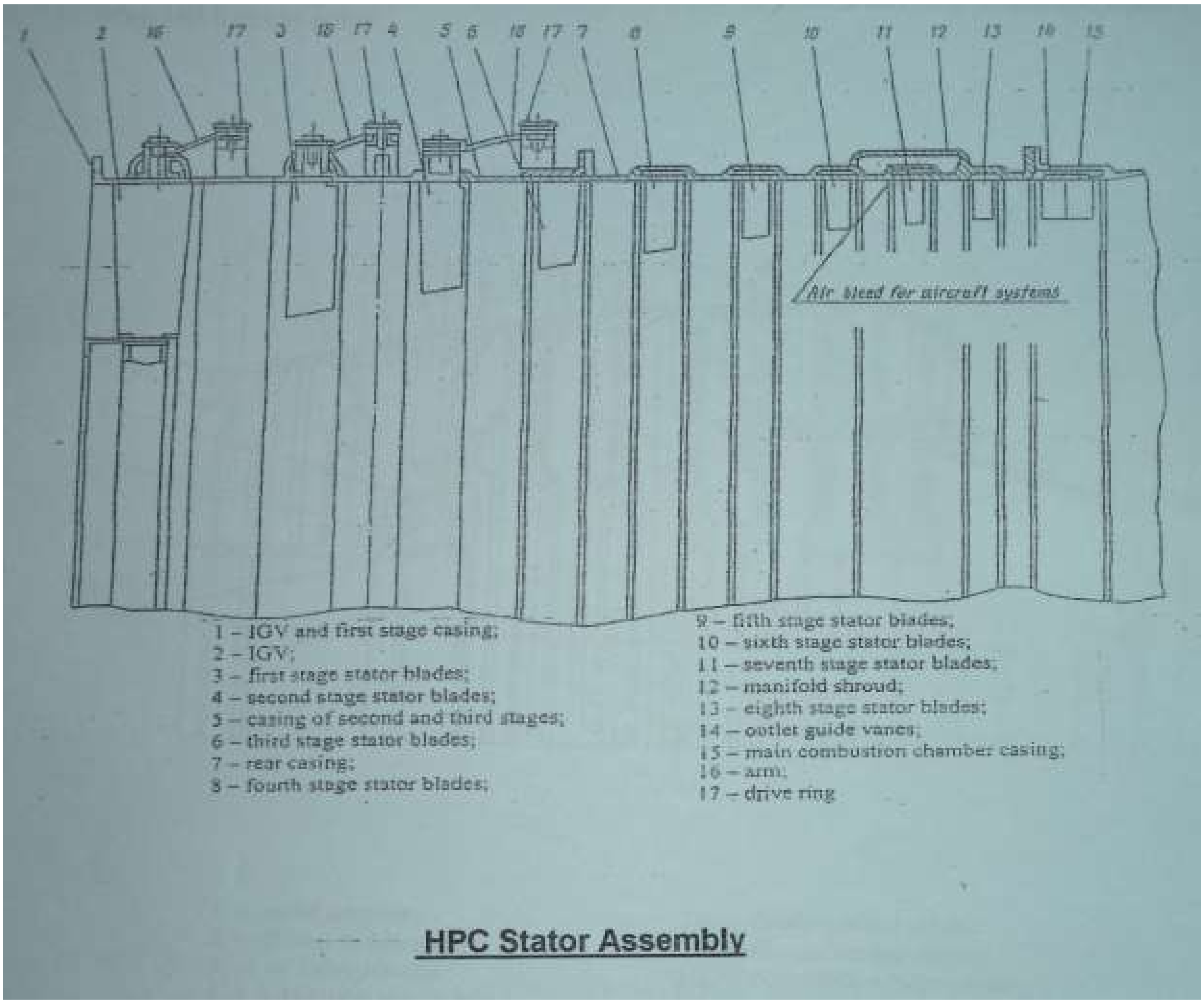
The compressor blades are made up of titanium alloys, which can sustain high pressure due to their material composition. The raw material of titanium alloy is forged for different size to get the shape of blades and discs are made from high speed carbide steel according to their stage. After forging these blades will undergo series of operations like profile making by CNC machines, broaching, milling, grinding, polishing, coatings are carried out on raw materials in order to have the desired shape of blades and discs.

High pressure compressor:

High pressure compressor is 9 stage axial compressor which is having a compression ratio of 6.6:1 .In HPC all the rotor blades are not of variable type due to their high rpm but the first 3 stages of HPC stator blades are having variable angle of attack so they can guide the inlet air of the HPC.HPC is driven HPT by means of HP shaft and this entire mechanical linkage is said to be HP spool.

The first 1stage of HPC blades are made of titanium alloy, next 8 stages blades are made up of nickel based alloy in order to sustain the heat radiated from the combustion chamber. All the compressor discs are made up steel alloy in order to protect from the heat radiated from combustion chamber there is a special disc named Labyrinth which is made of nickel based alloys and this comes between the combustion chamber and compressor.

The HPC blades are also forged blades so they also have to undergo series profile making operations except first stage remaining stages undergo cold rolling process



HPC Stator Assembly

due to their shorter length when compared to remaining blades. After cold rolling the blades are straightened, trimmed and polished.

Special parameters:

LPC compression ratio 3.5:1 HPC compression ratio 6.6:1

Total Compression ratio 23:1

Intermediate casing:

Intermediate casing section which is used to support the HPC casing and LPC casing and this is used to hang the engine to aircraft. It is made up nickel based alloy. At intermediate casing there is bypass of LPC compressed air with bypass ratio of 0.57:1 .The bypassed air is used in cooling process by exchanging heat at combustion chamber through heat exchangers.

Central bevel drive which is said to be heart of power transmission in engine is also placed in this intermediate casing. Aircraft gear box transmits power to engine gear box through means of flexible shaft and engine gear box transmits rotational power to central bevel drive and the bevel gear in this transmits power to HPC rotor in changing 90 degree of direction of power. So when the HPC rotor attains 40% of its rpm automatically the aircraft auxiliary power cuts off and engine turbine power comes into act.

While the power is generated by turbine some amount of power is transmitted to aircraft gear box from central bevel drive in opposite direction in order to run the aircraft accessories.

- In RD 33only intermediate casing is present with a bypass ratio of 0.45:1

COMBUSTION CHAMBER SECTION

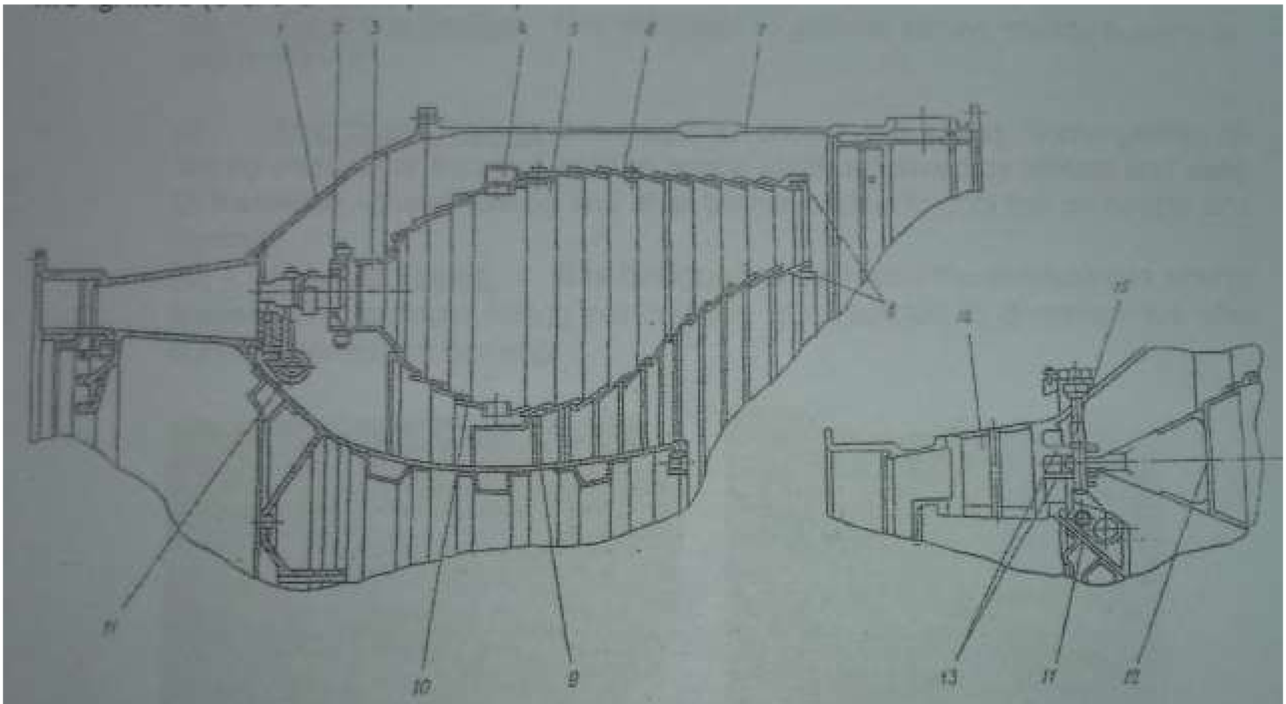
In AL31FP engine the combustion chamber is annular type combustion chamber which is having 28 atomizers with two manifolds. It is made of 5 components namely outer casing, inner casing, flame tube, manifold, burner which are forged separately and assembled together each other.

As the combustion chamber is very hottest section in chamber, so it is made of Nickel based alloy called BT-20 and the casings are made up isothermal forging. The forged casings,manifolds,flame tube, burner, atomizers are sent to electron discharge machine in order to place holes on them as there is no standard drill bit. EDM is the efficient process because of drilling of holes takes more time.

After combustion chamber there is an air heat exchanger which is said to be turbine cooling section. It is the same idea as fins on external surface of motorbike engine. It is activated only when the exhaust temperature of combustion chamber increases by 800 degree and as well as the rpm of high pressure shaft goes to 92% of its initial rpm.

Air heat exchanger is combination of series of small tubes bent together and braced to each other. One side there is an inlet of hot gases from combustion chamber and other side there is an outlet for cooled hot gases to HPT nozzle guide vanes (1 stage turbine stator blades).

- In RD 33& R29B the combustion chamber is annular type.
- In R25 U the combustion chamber is can-annular type.



1 – outer casing;
2 – vane swirler;
3 – vortex chamber;
4 – air intake;
5 – flame tube;
6 – mixing holes;
7 – heat exchanger casing;
8 – flanges;
9 – inner casing;
10 – annular slot;
11 – fuel manifold;
12 – annular shroud;
13 – brackets;
14 – strut;
15 – radial pin

Combustion Chamber.

TURBINE SECTION

In AL31FP the turbine section is a two stage impulse reaction type of turbine. One stage is High Pressure Turbine (HPT) and the second stage is Low Pressure Turbine (LPT).

The turbine discs and blades are made up of nickel based alloys which are capable of heat resistant. The blades are specially made by investment casting process with no post surface finishing process because of its precision.

The HPT turbine is connected to HPC by means of high pressure spool and LPT and LPC are connected by means of low pressure spool. The low pressure spool will pass through hollow structure of the low pressure spool.

As the hot gases coming from the combustion chamber initially the gases is guided by the turbine stator blades said to be 1st stage turbine nozzle guide vanes and passed over through 1st stage turbine rotor blades and next it passed over the next stage of blades. As the gases at stator blades they are guided and pressurized. The gas molecules expand through the rotor blades by virtue of kinetic energy of gas molecules transfer to the blades, so there be an impulse reaction develops over the blades and it causes the turbine to rotate.

There is special cooling unit said to be turbine cooling unit which is activated at high temperatures which allows to extend the life period turbine section. The HPT rotor blades of is having a typical aero foil shape with small holes drilled by electron discharge machine so that these holes are interconnected with disc so that there is cold compressed air flow transfer between the compressor and turbine through bleed air system.

- In RD 33, R25U, R29B are also having turbine of two stages one LPT & other is HPT.In these also there turbine cooling system, varying by design. These engine turbine blades are hollow and a core is present in them and compressed cold air is passed over the core.

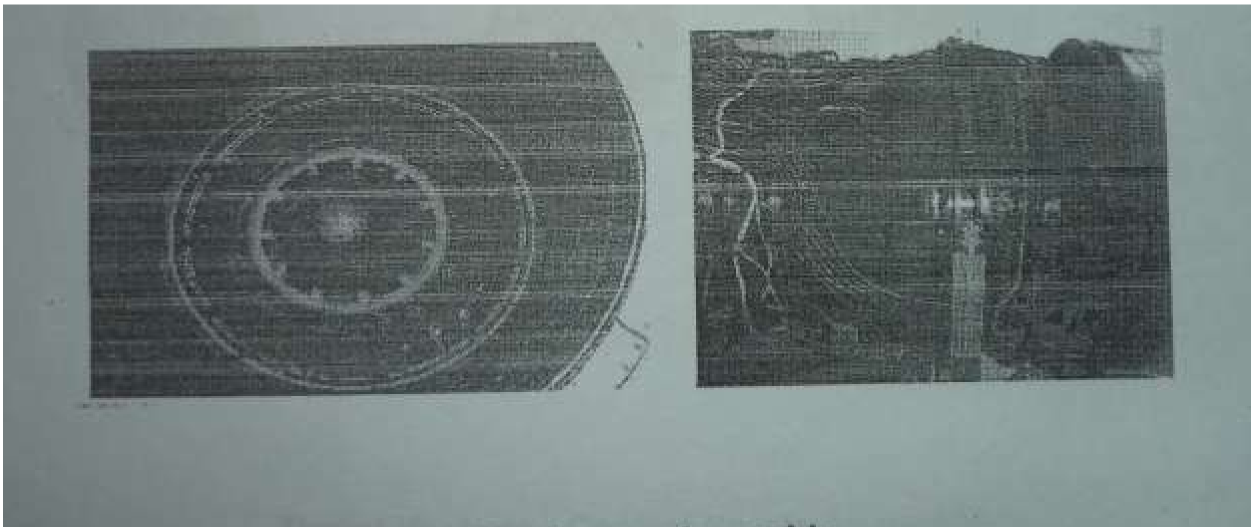
DIFFUSER & AFTER BURNER SECTION

In **AL31FP** the afterburner in having two sections that is diffuser section and afterburner manifold section.

Diffuser section:

Diffuser section is the section, the hot expanded gases coming out from the turbine is slowed downed by using diverging duct so that the gas is pressurized and velocity is decreased. So that the re-pressurized gas is supplied to after burner manifolds. There are two casings one in inner casing and outer casing made up nymonic, and compressed air from compressor is passed between casings in order to cool the hot gases by using bleed air system.

Afterburner section:



After burner Assembly

Afterburner section consists of 5 fuel manifolds out which primary manifold supplies fuel to the flame tube and remaining manifolds spray the fuel. At initial point of afterburner activation, fuel is supplied to primary manifold and oxygen is supplied for burning initializing process. Then the after burner fuel pump supplies fuel to 2,3,4,5 manifolds in sequence. Once all the manifolds have fuel there is flame carrier and burning starts at all manifolds and oxygen supply cuts. There are two flame stabilizers in order to stabilize the flame one big and the other is small. By activating the afterburner there is increase 6-9% of actual thrust and there is increase of 15 degree temperature at every 5 secs.

Parameters:

Normal exhaust temperature 460-650 degree.

Normal thrust generated 12500 kg

Afterburner thrust generated 13250-13500 kg

- In RD 33,R25U,R29B are having afterburner sections

NOZZLE SECTION

In **AL 31 FP** turbojet engine the nozzle is converging and diverging duct with variable thrust controller, tilting to angle of 14 degree up and down in vertical motion to engine axis.

Major components and their function:

1) Jet nozzle casing: It is the outer casing which is used to connect the diffuser section and nozzle section. Made up of titanium based alloy for strain hardening.

2) **Flaps:** Flaps are the moving parts in nozzle section which plays a vital role in controlling the throat area of nozzle. There are different flaps in nozzle section while viewing in top view with reference to inlet of the engine.

I) Inner flaps/Lower flaps are the flaps before the throat of the nozzle are made of nickel alloy because of high temperature contact.

ii) Upper flaps are the flaps after the throat area.

iii) Outer flaps are the flaps covering the inner components of hydraulic, pneumatic and mechanical linkages and forming nozzle shape

3) Spacers: Spacers are the stationary parts and acts as supporters to the flaps which are interconnected with hinge supports. There also different types as related to the flaps and supports respective flaps.

*In alignment one flap and one spacer are arranged in circular shape and connected with hinge supports.

4) Mechanical linkages: Mechanical linkages are the connecters for different components of and used in mechanical movements with the activation of pressure cylinders. Linkages like crossbeam, lever, telescopic support, brackets, piston rods, limiters.

5) Hydraulic cylinders: 2 types of hydraulic cylinders are used in nozzle section.

i) One type consists of 4 cylinders which are used to tilt the nozzle in vertical direction up and down 14 degree with reference of front view.

ii) Second type consists of 16 cylinders which are used in controlling the throat area either expansion/contraction with mechanical linkage to the flaps.

6) Pneumatic cylinders: There are 16 pneumatic cylinders which are used for restricting the over expansion/contraction of nozzle throat area by flaps and it is limited by limiters around the upper flaps inter connected with cylinders.

7) Graphite ring: Graphite ring is a ring which is fitted in between the nozzle and diffuser section of afterburner and provides lubrication during the time of tilting and covered with jet nozzle casing.

Special coating on the components:

As the nozzle section is hot part in the engine and due to the mechanical movement of the nozzle there will destruction of parts due to heat and wear stress. In order to avoid the destruction some special coatings are carried on some parts which are continuously exposed to heat and stress.

1) Yttrium stabilized zincronia coating is used as heat resistant and it is done on flaps and spacers.

2) Molybdenum coating is used as wear resistant and it is done on pivot joints, thrust bushes.

3) Graphite coating is done on graphite ring in order to maintain the lubrication at high temperatures generated due to continuous tilting.

4) Nickel-aluminum coating is for anti-friction and heat resistance done on graphite ring before graphite coating.

5) Detonation coating is tungsten carbide based coating for wear resistance done flaps hinge joints.

Description about Nozzle:

The nozzle is thrust variable controllable by expanding/contracting the throat area and changing the direction of nozzle by tilting vertically up and down at an angle of 14 degree. The nozzle is aligned 32 degree 44 min to the vertical axis of engine opposite to the side of engine position in aircraft.

Ex: If the engine is left side of aircraft the nozzle is placed 32 degree 44 min right to the engine axis.

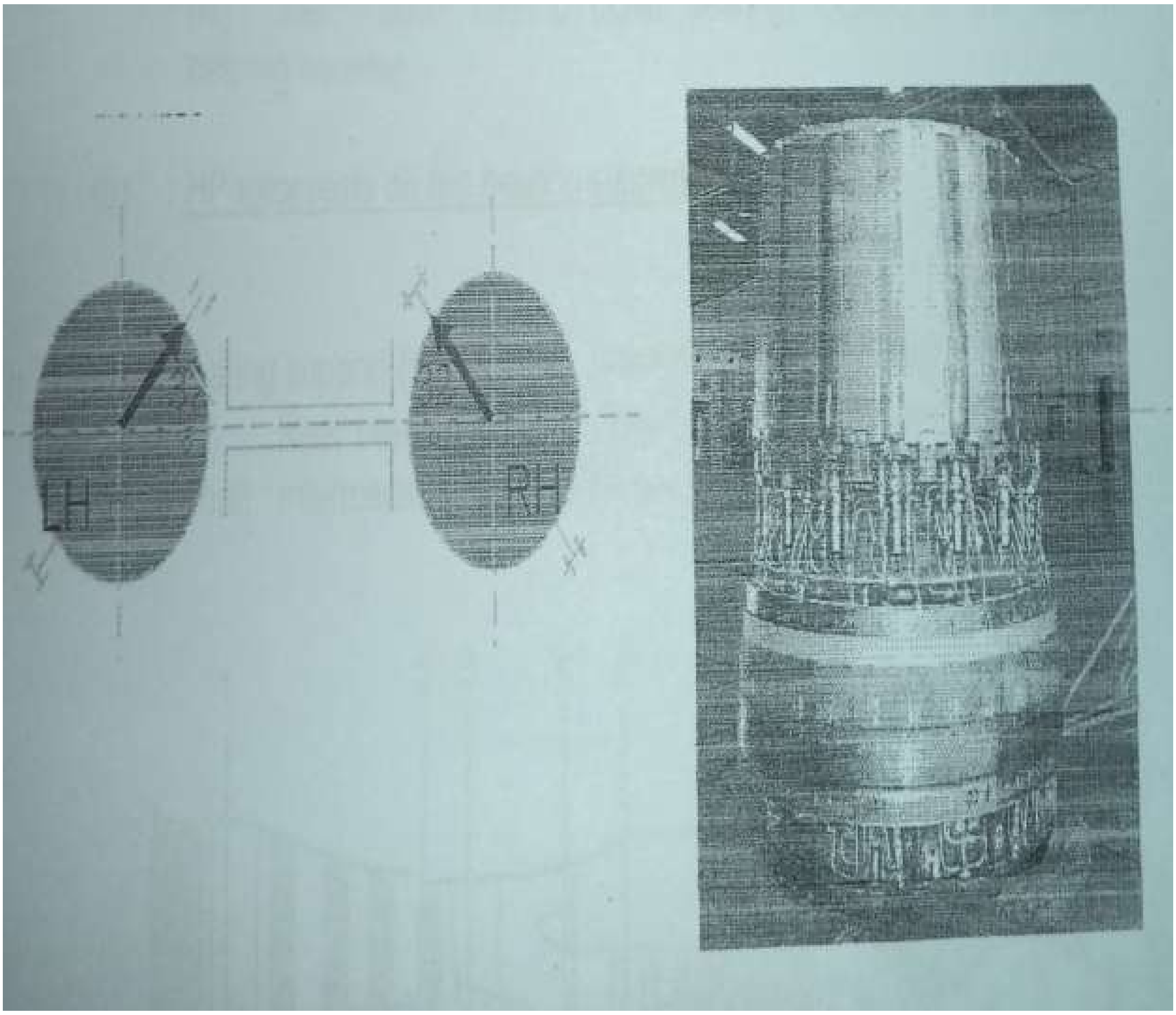
Standard parameters:

*Thrust generated 12,500 kg at full reheat time/combat mode.

*Air mass flow 112 kg/sec.

*Maximum temperature 895 degree.

- In RD 33 the nozzle can contract and expand but it cannot tilt 14 degree.
- In R29 B and R 25U the nozzle are not of variable type.



GEAR BOX

Gear box is a type of mechanical device which is used to transmit power to various components through a single input power at desired rpm. In gearbox rpm is increased/decreased according to design.

Gear box are manufactured by gear shop mainly in three sections

A) Gear casings

b) Gears & shafts

c) Small parts.

All section parts are manufactured from magnesium alloys. The raw material is forged and sent to the gear shop. The raw material undergo different manufacturing process in order to obtain desired shape. It undergoes ion nitrating process in order to obtain hardness 65-90 with respective component.

Special process

Hard alloy Graphite coating

Process is carried out to increase the wear resistance of non-lubricated sliding/rubbing/mating parts.

Hard alloy coating is a process for application of coating by electric spark method to increase the wear resistance. The component to be coated is held and rotated on 3 jaws chunk of the equipment and the special gun holding electrode of hard alloy T15K6 is guided manually on the surface of part to generate the spark to facilitate material deposition.

In graphite coating graphite electrode is used. It is the process of depositing the lubricating layer.

In **AL31FP** there are 2 gear boxes named

1) Engine gear box (EGB) -KDA (Russian name)

2) Aircraft gear box (AGB)-BKA (Russian name)

Initially at starting of engine a turbo starter an initially power plant, drive through battery (DC motor), so the power is transmitted from turbo starter to AGB through quill shaft and from AGB the power is supplied to EGB through means of flexible shaft. EGB transmits the power to CBD and CBD to HPR so that this cycle continues for 40 secs from turbo starter ON and automatically cuts off when engine becomes power plant. It will become reverse cycle for transmitting mechanical power to electrical power through dynamo to run all the cabin and electrical modules.

Engine gear box

EGB is the one of the engine component which drives all the engine accessories through alternative two way power supply either from AGB or Central bevel drive (CBD).If power supplied from AGB then CBD comes as an output, if power transmits through CBD then AGB comes as an output.

EGB is having 10 gears to transmit power and it requires 20 bearings to support the both sides of gears.

Main driving units of EGB

1) Main fuel pump.

2) NP pump -regulates nozzle diameter.

3) Booster pump-It boosts fuel by pressurizing the fuel to 5-8 kg/cm^2.

4) Afterburner fuel pump-It supplies fuel to after burner fuel manifolds.

5) Oil aggregated unit-It supplies fuel to AGB & EGB and receives it from them and sends back to oil tank through oil filter.

6) Centrifugal breather-It removes the air from the oil in order to supply uniform oil for lubricating purpose.

7) Either AGB or CBD.

* In RD 33, R25u&R29B also there is AGB and CBD.

Aircraft gear box

As SU30MKI aircraft is powered by twin engine, a single turbo starter cannot alone starts the engines so an additional gear box is designed said to aircraft gear box for each engine. So turbo starter transmits power to both of the engines.

AGB is the aircraft module which drives all the aircraft mounted accessories from the alternative power supply from either turbo starter/EGB. If power supplied from turbo starter the EGB is an output, if power supplied from EGB the turbo starter cuts off and dynamo activates for supplying electrical power to pilots cabin and battery charge.

AGB is having 17 gears supported by 34 bearings.

Main driving units of AGB

1) Booster pump supplies fuel to engine components as lubricating purpose.

2) NP-128 pump is a hydraulic pump supports landing gear systems.

3)Scavenging pumps is of oil and fuel type which receives the oil/fuel from the respective modules and sends to oil/fuel filters and sent to respective tanks.

4) D3M tachometer is a mechanically rpm measuring unit by developing magnetic flux.

5) Booster pump-Booster pump supplies lubricating fuel to engine mounted systems.

6) Turbo starter/EGB.

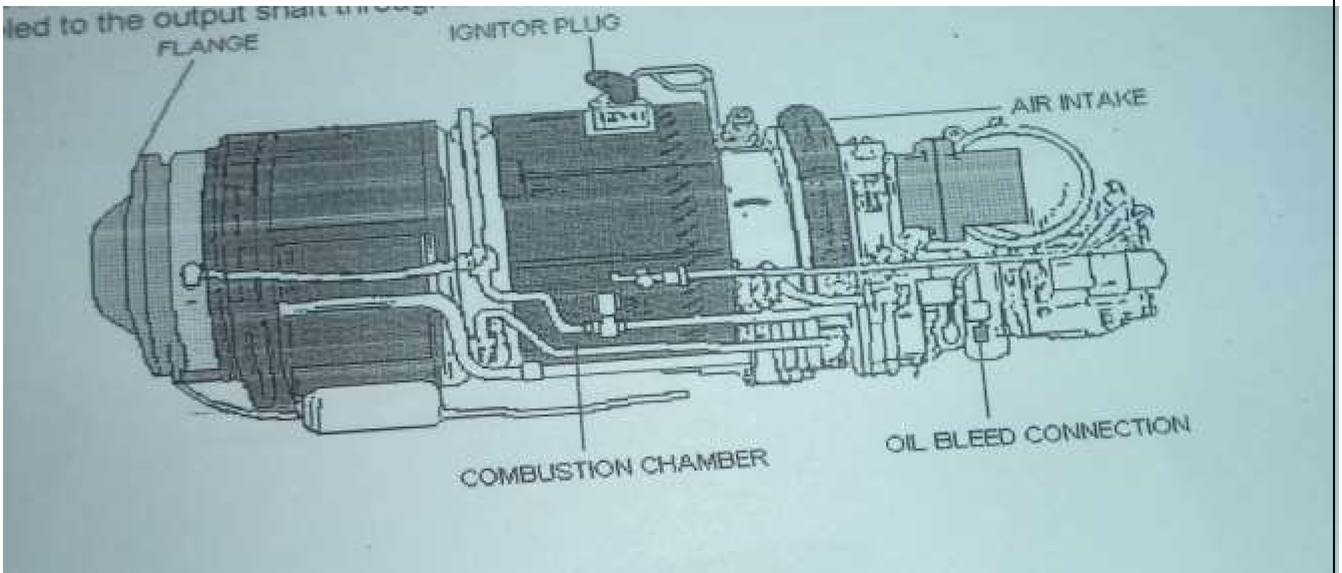
- In RD33 AGB is present as the aircraft is powered by twin engine.
- In R29B & R25 U No AGB is used because of single engine powered aircrafts and onboard system runs on turbo generator which works on same principle of turbo generator.

TURBOSTARTER

Turbo starter is turbine powered small engine which works on the small thermodynamic cycle of jet engine but of smaller in size as per its power generation. A single turbo starter is used in the aircraft.

The turbo starter is initially powered by battery so that the centrifugal compressor in the starter starts and sucks the air and small amount fuel is sprayed through atomizers of manifold and spark is introduced. The combustion takes place and the gases start expand through turbine. The power is transmitted to both aircraft gear boxes of engine parallel through means of quill shaft.

- In RD 33 turbo starter is used in the aircraft which is powered by twin engine.
- In R29B & R25U turbo generator is used, as there is no aircraft gear box. All the aircraft accessories are powered by turbo generator by means of reverse power transmitted by turbine. The only difference between turbo starter and turbo generator is turbo generator can transmits power in two directions i.e. it cans supply and receive power like a dynamo.



Turbo starter (Left View)

Major parts of AL31FP & their functions

1) Inlet Guide Vanes-These vanes act as diffuser and directs the intake air, these vanes are variable with hydraulic system and they have local angle of attack -30 to 0 degree.

2) LPC- Low Pressure Compressor is of four stages axial type which is supported by roller bearing at front and at rear by ball bearing and there is compressor ratio of 3.5:1

3) Intermediate Casing-This casing act as a structural member between LPC, HPC and aircraft and carries primary load. There is also a bypass duct which bypass the LPC compressed air with bypass ratio of 0.57:1.The bypassed air is used as coolant for combustion chamber through air heat exchanger and afterburner.

4) HPC- High Pressure Compressor is of 9 stages axial type and having 1, 2,3 stage stator with variable control having angle -25 to 0 degree and it is supported by ball bearing at front and rear by roller bearing.

5) Combustion Chamber- Annular type having outer, inner casing with flame tube and manifolds for fuel supply.

6) Turbine Inlet guide Vanes-These vanes are used to guide the high pressure gas coming from combustion chamber and these vanes are outer cased with air heat exchanger pipes.

7) Low Pressure Turbine- LPT is single stage turbine with impulse and reaction type blades mounting on it.

8) High Pressure Turbine-HPT is also a single stage turbine with impulse and reaction types blades mounting on it.

9) Nozzle Diffuser-This diffuser acts as to increase the pressure and decrease the velocity of the expanded gases for helping in accurate combustion processes in afterburner.

10) Afterburner-After burner is the device which is used for thrust augmentation which is approximately operated for maximum 30 secs at the time of takeoff or combat mode because of lot of consumption of fuel.

11) Nozzle-This Nozzle thrust vectoring type which is moved up or down at an angle 14 degree and this is fitted to engine axis at angle of 32 degree and mounted with 2 degree of tolerance and this based on the side of engine. If it is left the engine is fitted to 32 degree right to the engine

12) LPR- Low Pressure Shaft is the shaft which is used to connect the LPC and LPT.

13) HPR-High Pressure Shaft is the shaft which is used to connect the HPC and HPT.

14) Engine gear box- EGB is fitted on the engine and used to drive the engine accessories like main fuel pump, auxiliary fuel pump, after burner fuel pump, High pressure pump, centrifugal breather and oil pump

15) Aircraft gear box -AGB is placed on aircraft engine strut so that it is mechanically connected to EGB through flexible shaft and drives EGB with help of turbine generator at the time of starting of engine.

16) Aggregates-There are 29 repairable aggregates and total aggregates are around 48.Remaining aggregates except 29 are not repairable, so after overhauling they will go for the functional test and used further if it is working properly or it is rejected if it doesn't work.All these 29 repairable aggregates used in the engine system out of these 12 are electrical and 17 are mechanical

Mechanical aggregates

a) Fuel Oil Distributor-In this engine pump named RT-31 VT1 manufactured by ADL and used to distributes fuel between the first and second stage fuel manifolds of main combustion chamber.

B) Distributor of after burner -Pump named RTF 31AT1 manufactured by ADL distributes fuel flow between 5 manifolds and maintains minimum fuel flow in after burner manifolds.

c) Booster pump-Pump named DTsN82 manufactured by ADL which increases fuel pressure from aircraft fuel tank to fed high pressure fuel system and supplies fuel to main fuel pump NR 31BT1 and afterburner pump FN 31 AT1 and plunger pump NP1600.

d) Emergency drain unit-Drains fuel from aircraft fuel tanks at emergency landings and operated from cockpit.

e) Plunger pump-Pump named NP160D manufactured by ADL which supplies fuel to hydraulic cylinders through jet nozzle and reheat controller RST 31BT1 for control of jet nozzle throat area and supplies fuel to electro hydraulic distributor EGR4BT1 for controlling tilting of thrust vectoring system.

f) Fuel oil filter-Filter of tolerance with 16 micron to filter the fuel fed system engine units from foreign particles.

g) Regulator of nozzle and afterburner-Pump named RSF 31 VT1 which controls3 parts

i) Jet nozzle control system for controlling the throat area for thrust control according to engine throttle position.

ii) After burner control system controls fuel supply to afterburner manifolds through afterburner fuel distributor RTF31AT1 when afterburner is engaged.

iii) Turbine cooling control system gives hydraulic command for switching off and on of the system.

h) Regulatory pump-Pump named NR31VT1 which regulates 2 units

i)Pumping unit increases fuel pressure supplied from booster pump in relation to high pressure rotor RPM.

ii) Regulator unit -controls fuel supply to main fuel distributor RT31BT1 in relation to HP rotor RPM, inlet air temperature,atmospheric pressure.It also controls fuel supply to main fuel supply to the hydraulic cylinders of LPC inlet guide vanes and high pressure compressor variable stator vanes in relation to inlet air temperature and HP rotor rpm

i)Reheating pump- Centrifugal Pump named FN31AT1 manufactured by ADL intended for increasing the fuel supply to afterburner fuel distributor RTF31AT1 through jet nozzle and afterburner fuel distributor (RTF 31AT1) through jet nozzle and after burner regulator (RSF31BT1).

j) Gas couple type thermo sensor-It senses inlet air temperature (t1) and generates hydraulic command to main fuel pump (NR-31BT1) and jet nozzle reheat regulator (RSF31 BT1) for correction of fuel flow.

k) Plunger pump control unit -It controls the outlet pressure of plunger pump (NP160D) for actuation jet nozzle hydraulic cylinders.

l) Aggregate for setting P1 turbine-controls turbine expansion ratio p2/p4 by changing jet nozzle thrust area.

m) Safety valve (KP96VTI)-It by passes the fuel when plunger pump (NP-160D) outlet pressure increases beyond 240kg/cm2.

n) Fuel feed meter-It is meant for after burner fuel ignition through hot streak method.

0) Electro hydraulic fuel distributer (PMZ VOSKHOD EGR-4VT1)-It distributes fuel to hydraulic cylinders for tilting of jet nozzle through 2 independent electrical channels one hydraulically channel

p)Fuel oil-heat exchanger(TEPLOBMENNIK 6139T)-It is cross flow heat exchanger which cools the heated oil by using the outlet fuel from and afterburner fuel system

Electrical aggregates

a)Sensor of tachometer(ELECTROPRIBOR D-3M)- Aggregates is intended for electrical signal with frequency proportional to that of LP rotor Rpm(n1) and HP rotor RPM(n2) display at cockpit.

b)Synchronous transmitter(PRIBOR DS-11V)-3no's are installed on the engine it is based on principal of synchronous measures positions of-flaps in jet nozzle,LPCIGV,HPCVSV and signal to MFWS.

c) Sensor of lubricate oil level indicator (TEKHPRIBOR OSMK 8A 47)-oil level sensor is mounted on oil tank measures the level of engine oil and provides electrical signal to aircraft electrical system.

d) Vibrio speed sensor (MV271G)-Measures vibration of casing. The unit converts horizontal/vertical vibration in to electrical signal and supplies to complex regulator of engine (KRD-998).

e)Gas turbine engine power unit -(KRASNY OKTYABR GTDE117-1MO)-It is intended to rotate engine up to self-sustaining RPM during engine starting on ground and up to cranking RPM during cold/wet cranking and preservation/DE preservation.

f) Electromagnetic valve-(SEPO-ZEM MKV-253A)-This is solenoid valve meant for anti-icing system. It receives feedback from ice detection unit of aircraft and automatically opens the airline tapped of 7th stage HPC to be supplied IGV in the event of ice forming

g) Complex regulator engine (SEPO-ZEM KRD99B)-It is electronic control unit which controls/regulates fuel flow throughout all engine modules and rating. In addition it receives signals from feedback sensorsDT211, P109, MV271G, DP110, DAT4, T-142, SPT88.22 and provides command for control of engine starting, afterburner mode, turbine cooling,surge,LPCIGV,HPCVSV angle,LP HP rotor RPM,turbine exit temperatureT4,I t also sends signal to flight data records (FDR) MFWS-multi functional warning system&VIRS-voice information recording system.

h**) Movement sensor** (KPD-99B) DP110-There are two no's (DL-110) on the AL-31FP engine. They work in the principal of RVDT-rotation variable differential transducer).They monitor (LPCIGV) flaps position&HPC-VSV positions and provide signal to KRD-99B.

i) RPM sensor(Dchv-2500A)-The aggregate is a pick up sensor which measures LP RPM(n1)&HP RPM(n2).There are 5 DChV-2500A on engine. Three of them are installed in LP transducer gear box and they monitor n1 and other two are installed in A/C gear box (BKA-99MK) and monitor n2.

j) Heat resistant pressure indicator MST-6.5A-It is pressure switch which controls selects the correct air pressure line to be fed to engine accessory gear box bearings.

k)Heat resistant vibration stability pressure indicator MSTV1-It senses the pressure in oil supply line of turbo starter(GTDE-117-1MO) and automatically stops turbo starter operation in case of oil pressure is less than 1 kgf/cm^2.

l)Sensor of transformer pressure(DAT-250)-It is a pressure sensor which measures the pressure of fuel in plunger pump(NP-160D) pressure line and transforms this in to electrical signal.

m) Transformer pressure sensor(DAT4)-It is a transformer type pressure sensor which measures pressure of oil line and transforms this in to electrical signal to feed engine combined controller.

MANUFACTURING PROCESS

Foundry shop is the base shop from where the manufacturing process starts. It is divided into two sections of manufacturing one is forge section and casting section

Forge section

Forging is a type of manufacturing process in which the material is heated to plastic state and pressed with dies by applying heavy force.

"Forge is a manufacturing process involving the shaping of metal using localized compressive forces"

Steps for forging

a) Billet preparation- Billet is prepared for raw materials of particular dimensions as per requirement given in technology

b) Upsetting-Upsetting is a reshaping forge process of localized increase cross sectional area of billet.

It can be done by mechanical load and as well as by electric machine by induction heating within equipment.

c) Extrusion-A pre shaping process used to create objects of a fixed cross-sectional profile by restricting the flow in other direction. A material is pushed or drawn through a die of the desired cross-section

Initially the billet is dipped in hot salt water bath and the upper portion of billet held and pressed where as he lower portion is held and elongated.

d) Heating-Now the reshaped billet is heated to around 900-1400 degree Celsius according metallic composition.

e) Forging-The red hot billet is now pressed according to different types of forging press available in the shop.

Different types of forging

Free forging-This is open forging process of shaping metal using flat discs.

Hammer forging-A process of shaping hot metal completely within cavities of 2 dies using hammers.

Envelope forging-A closed die forging process with machine allowance of

1.5mm-3mm using press hammers.

Precision forging-Near net shape with least or no machine allowances.

Special forging process

Isothermal forging-Pressing dies are maintained at same temperature of work piece temperature which is capable of producing near net shapes.

The temperature is constant for both the component & dye. For this the component(shape has to be changed) is kept in contact with the dye and allowed for 5 minutes to get heated until the temperature of component becomes equal with the dye and forming is done to get the required shape.

Pneumo thermo forming-This process is used for forming diffuser casing of engine and spherical casing of jet nozzle.

Process-This is a super plastic forming process for achieving intricate contours in titanium alloys. The material to be formed is taken to super plastic temperature range of 1050 degree Celsius in argon atmosphere and then expanded by applying pneumatic pressure with argon. This process is used for forming of diffuser casing and spherical casing of jet nozzle.

Induction heating: Component loaded in Lathe machine and heated to 900 degree Celsius by induction heating method revolving at a low speed of 18rpm when component is red hot a roller of corresponding diameter is pressed into the component and rotated thus a concave/convex profile is formed. To avoid crack in sensitive material like BT-20 the part is preheated to 900 degree Celsius and rolled to concave/convex.

CASTING SECTION

Casting is a manufacturing process by which a liquid material is usually poured into a mold, which contains a hollow cavity of the desired shape, and then allowed to solidify. The solidified part is also known as a casting, which is ejected or broken out of the mold to complete the process

Sand casting-Sand casting is a done with sand dies made from drag (bottom section of die) and cope (upper section of die) and material is poured into die through the pourer and observe red through riser. Sand casting is a very low expensive process but very less precision is obtained. Generally components are prepared from two metallic alloys one is aluminum alloy and magnesium alloy.

For Aluminum

Fresh Silica sand with used silica sand (1:3) ratio is used to prepare sand mixture + dextrium + distill water.

For Magnesium

The above composition +BM mixture + sulphur +Boric acid. Because in order to stop oxidation process on open to atmosphere.

Investment casting-Investment casting is very costly process with high precision almost zero error machine process. It is done by ceramic casting/urea casting. It is said investment because the mold wax cannot be re-used.

Steps in investment casting:

a) Core mass preparation

By urea

Hydrate ethyl slicate+spirit+distill water+HCL+acetic acid+Sodium hydroxide+industrial urea.

By Ceramic powder

Hydraite ethyl slicate+spirit+distill water+HCL+acetic acid+Sodium hydroxide+cermaic powder

By Ceramic powder for single crystal-

Quartz (pure aluminum) +tiO.2+electro corundum powder

By Ceramic powder for Equayside -

Electro corundum powder silicon oxide.

b) Core inject from die

The core mass is molten state and forcibly injected to die through hydraulic machines and left for 3 days to solidify.

c) Levelling of core

The solidified core is physical observed and uniformity core distribution is checked and if required some core mass is painted if observe red it is less or polished if observed it is more.

d) Centering of core

The leveled core is centered in ammonia powder and maintained at 1000 degree Celsius for 10 hours. It is done to increase the strength of the core.

E) Pattern making & assembly

In order to place the core and hold the core wax patterns are made for different shape of core and core is assembled in wax and held in wax pattern.

F) Ceramic coating

The totally assembly is subjected to coating of ceramic paint and dried in ammonia chamber. The coating goes is layer by layer and once requires thickness layer is reached it is subjected to next step.

G) De waxing

The wax is removed by heating the coated assembly to around 150 degree Celsius for 2-3 hours.

H) Calcination & metal pouring (in vacuum)

Calcination is a preprocess of metal pouring in order to make the detained assembly to sustain the metal temperature so it is also heated around the molten metal temperature and then metal is poured through pourer and observed through riser.

For single crystal there is special process in which before metal pouring a metallic grain seed is placed with tungsten helix which makes a single crystal gets contact with molten metal and a single crystal is developed on cooling

I) Cooling

After vacuum metal pouring the assembly is subjected to cooling. Cooling general takes place in three ways

i) Multi directional cooling takes place normally when it is left in atmospheric temperature after pouring and multi crystals develops and which are responsible for cracks on exposing to high strain.

ii) Directional solidification takes place when cooling is done from one side and the crystal develops from the cooling side and develops grain is unidirectional and highly resistive in that direction.

iii)Single crystal cooling in this when the single grain seed and comes in contact with molten metal cooling takes place in unidirectional and the total component forms a single crystal and which are high strength.

j) Knock out and cutting

The ceramic coat is removed by knocking the cooled assembly with hammers. After removal of ceramic coat the excess metal is removed by cutting.

K) Leaching

The components are separated from assembly and the components are placed in KFl2 &KFl4 alkyl solution and heated so that central core is removed.

So final the component is obtained with almost zero machine allowances.

SHEET METAL SECTION

The manufacturing process involved in this shop is mainly refers to sheet metal components.

Pressing-pressing is done by either mechanically or hydraulic methods. It is a machine that changes the shape of a work piece by the application of pressure

This is of two types non cutting & cutting.

Non cutting involves only change in shape by stretching, forming (change of shape in the multiple axis), and bending (change of shape in single axis) but don't involve in reducing the metal.

Cutting involves in reduction of the material by trimming, blanking (initial shape is cut for further operations & useful part is more than crab), piercing (holes can be made of any shape), notching, punching (circular holes can made).

Shearing, also known as **die cutting**, is a process which cuts stock without the formation of chips or the use of burning or melting.

Blanking: A blanking die produces a flat piece of material by cutting the desired shape in one operation. The finish part is referred to as a blank. Generally a blanking die may only cut the outside contour of a part, often used for parts with no internal features

Piercing: The **piercing** operation is used to pierce holes in stampings

Blanking and **piercing** are shearing processes in which a punch and die are used to modify webs. The tooling and processes are the same between the two, only the terminology is different: in blanking the punched out piece is used and called a blank; in piercing the punched out piece is scrap. The process for parts manufactured simultaneously with both techniques is often termed 'pierce and blank'. An alternative name of piercing is punching.

Cold rolling: This process is special for HPC last 4 stages blades due to their shorter length. This is a cost effective one and also fastest process but due to design of machinery, it can do for only shorter length blades.

In this process blades rolled between 2 rollers at room temperature for a period of 2-5 min depending upon the desired blade thickness which varies from 1.5 to 3mm.As the blades are made up of nickel based alloy there are very hard so nearly 2000KN force is applied and also a coolant is applied due to rapid heat generation. The coolant is special made up of 50% castor oil and 50% ethyl sprit.

MACHINING SECTION

Machine process is the next step of manufacturing process once component is ready from foundry shop.

1) **Curling**: The curling operation is used to roll the material into a curved shape. A door hinge is an example of a part created by a curling die.
2) **Groove making:** This is a special process carried out for making diamond shape of groove on the out diameter of discs. Special turning lathe machine is used in this process.
3) **Horning**: A horning die provides an arbor or horn which the parts are place for secondary operations.
4) **Trimming**: Trimming dies cut away excess or unwanted irregular features from a part, they are usually the last operation performed.
5) **Notching** is a metal-cutting process used on sheet metal or thin bar stock, sometimes on angle sections or tube this is a precursor to some other process: such as bending a corner in sheet or joining two tubes at a tee joint, notching one to fit closely to the other
6) **Milling** is the machining process of using rotary cutters to remove material [1] from a work piece advancing (or feeding) in a direction at an angle with the axis of the tool.
7) **Turning** is a machining process in which a cutting tool, typically a non-rotary tool bit, describes a helical tool path by moving more or less linearly while the work piece rotates.
8) **Drilling** is a cutting process that uses a drill bit to cut or enlarge a hole of circular cross-section in solid materials.
9) **Boring** is the process of enlarging a hole that has already been drilled(or cast), by means of a single-point cutting tool
10) **Reamer** is a type of rotary cutting tool used in metalworking. Precision reamers are designed to enlarge the size of a previously formed hole by a small amount but with a high degree of accuracy to leave smooth sides.
11) **Broaching** is a machining process that uses a toothed tool, called a **broach**, to remove material. There are two main types of broaching: linear and rotary. In linear broaching, which is the more common process, the broach is run linearly against a surface of the work piece to effect the cut. Linear broaches are used in a **broaching machine**, which is also sometimes shortened to broach. In rotary broaching, the broach is rotated and pressed into the

work piece to cut an axis symmetric shape. A rotary broach is used in a lathe or screw machine. In both processes the cut is performed in one pass of the broach, which makes it very efficient.

12) **Honing** is an abrasive machining process that produces a precision surface on a metalwork piece by scrubbing an abrasive stone against it along a controlled path. Honing is primarily used to improve the geometric form of a surface, but may also improve the surface texture.

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COATING SECTION

Coating section is mainly to coat different components according to their working section. Coatings are generally mean to increase the life of component by protecting the base material. It is done by avoiding the surface reduction like grinding of blade tips, coating on the winglets contact surface is to avoid the amplitude of vibration of LPCR blades first three stages.

There are different types of coatings here we mention only the process involve in the plant like Flame coating, Detonating coating, Plasma coating&MAP coating.

Sand blasting-Sand blasting is pre coating process done in order to remove the oil & grease drops from the surface of the components. Here the component are fixed in the chamber and sand is blasted (sprayed) on the surface of the component with high pressure so that oil & grease drops are removed. Sand size is small & thin there are four types of sand sprayed on surface depending on the component. Generally sand is of (white Al oxide grains).and blasting is done before the coating process for better sticking of coating by removing external foreign materials.

Flame coating-This is high temperature heat resisted coating on the component like flame tube, external mixture & stabilizer.

The coating is applied on the component by special gun through which BKNA material is fed in the form of cord. The spraying is done by melting the material with help of flame of oxygen& acetylene gases. The component is held at rotating chunk so that he uniform coating is ensured.

Detonating coating-This method is first introduced in India and involves coating of tips of wings for the three stages of compressor blades, spacer flaps of jet nozzle.

The powder material for coating are accelerated to high velocity and partially melted before it impinges on surface. Detonation is initiated in gas mixture near the closed end of barrel. The products of detonation, flying out from the barrel accelerated & heat the particles of powder up to partially melting. Coating takes places due to high speed of impact of particles in plastic condition. Process done in closed room because it crates sound due to supersonic speed of particles.

Plasma coating: This process is used for abradable coating on compressor casing ,antifriction coating on spherical ring an thermal barrier coating on hot end components and also for wear resistant.

Spraying of molten (or) heat softened material on to the surface. Here material is in the form of powder is injected in to very high plasma flame, where it is rapidly heated & accelerated to high velocity. The material impacts on the surface and rapidly cools forming coating. Materials used molybdenum coating, zirconium.

Map coating: High temperature heat resistant diffusion coating. Major component involved is HPTR blades.

This is Heat resistant coating, applied on the aero foil shape of the blade of HPT by high vacuum plasma method. Before coating aero foil blade are subjected to abrasive liquid process, polishing ensuring the surface finishing as per the etalon. Coating is done in vacuum with pressure and cooling of blades after completion of coating is caused out in chamber of spray coating up to a temperature of 150 degree Celsius. Thickness of coating is 0.035 to 0.045mm.After completion of coating annealing of blades are carried out vacuum with residual pressure not more than 10^-3mm of Hg.

WELDING SECTION

It is a process of joining of two similar metals by diffusion of any external or same metal

Electron beam welding-Used for welding of critical assemblies like compressor discs, casings advantage of this process is high welding penetration with les distortion welding under inert vacuum environment.

Here high velocity electron beam is ejected from a cathode & accelerated through lens to converge& hit the component thus producing enormous heat to force the metal and get joint.

This process is used for welding titanium based component like compressor discs casing in a vacuum (atmosphere) not less than (10^-5mm) of Hg.

This process is also used in drilling small holes which re smaller than the standard drill bit size by having cathode of same diameter.

Argon welding-It is the robotic welding system in argon chamber where there is a human entry with suit and required oxygen is supplied. For some components of welding man power is required and other parts are done automatic with robotic system.

This chamber has two robotic arms with 6 degrees of freedom it can move in 6 axis of rotation. One arm is for programming and other is to perform the job by using the program. Only Titanium alloys are welded loading in the presence of argon gases.

Brazing:

It is a process joining two dissimilar metal whereby a filler metal is heated above melting point and distributed between two or more close-fitting parts by capillary action& it's a semi-permanent joint.

Vacuum brazing with local heating: Used for brazing of hard alloys to tips of turbine blade. Here the major component involved is LPT blade.

This type of brazing is done with localized heating by arc in inert vacuum environment. The component to be brazed are put inside vacuum chamber and local heating is carried out with electric arc.

Vacuum brazing: Joining of parts by furnace brazing in vacuum. Here the components involved are labyrinth lower belt, upper labyrinth, flange with labyrinth, nozzle blade, LPT insert, nozzle vane of LPC.

Components cleaned with acetone & assembled in required fixture tightly by shoulder paste or shoulder strip such that gap between the component and fixture is not more than 0.03mm.Chromium oxide paste is to be applied on fixture to ensure non sticking of shoulder material to the fixture. Then the part is loaded in the furnaces after attaining the vacuum level of 10^-5bar.Heating is to be carried out 1060-1220 degree Celsius for 10-15 min.

Manifold Brazing with Rotation: Used for brazing of main manifold with rotation. Components involved are Main manifold with nozzle.

During furnace brazing there is a tendency for the braze material to flow down due to gravity which leads to unequal filling of braze material to avoid that the part is always rotated.

HEAT TREATMENT SECTION

Heat treatment is mechanically process by virtue of which the mechanical properties like strength, hardness, brittle nature, ductility of metals can be changed.

Generally when the surface of metals are under manufacturing process there is rigorous expansion and contraction takes place between atoms so there is some strain energy left in the metal, so in order to remove the strain in the metals some special heat treatment process is done according to composition of metal. In this process the atoms are suddenly heated to rise the atoms to higher energy levels and sudden cooling so that when atoms jumps down from higher energy levels to lower energy levels the energy released, so by this process the strain energy is also removed. But the nature of process depends on the metallic composition of component.

Ex: corrosion type of components the process are done at vacuum so there would no corrosion takes places.

Ion nitrating-This process is carried out to improve the wear resistant of gears, casings & shafts. Components involved are gears, rollers, quillshaft, shaft.

Ion nitrating is a high accuracy diffusion based process involving thermo chemical processing under vacuum level of 350pascal at temperature 550deg Celsius to 620 depending on material composition and utilizing the highly activated gas mixture of hydrogen & nitrogen in plasma state.

Chrome alitizing-Applied on inner casing of HPTR blade to improve high temperature oxidization, corrosion resistance. Component is HPTR blade.

Chrome alitizing is a process in which the components are heated at1030 degree celsius for 4 to 5 hrs. In vacuum atmosphere in presence of mixture of Al powder and chrome powder in closed packed condition to produce a diffused layer of chromium oxide.

Alphatising-This process is to improve wear resistance of bush, casing & pivot. Components are casing, case, pivot, bush, and piston.

It is a controlled oxidation process carried out on titanium alloys of BT-9 and BT-20 at a temperature of 850 degree celsius and a vacuum level of 0.008 to 0.004mm Hg for 6hrs to impart an oxide layer of titanium on finished components for achieving surface hardness.

Manufacturing of BKHA (Russian name) & BKpiH-5 (Russian name) cord-

BKHA elastic cord used as a bond coat&BKpiH-5 elastic cord used as a top coat applied on stabilizer & flame tube of AL-31FP by gas flame method to protect the parts from high temperature gas corrosion & erosion. Components are flame tube head, external mixture ,small & big stabilizer.

BKHA elastic cord is mixture of BKHA (piH75 23B) powder, rubber CKTH and catalyzer no.18 which is drawn in form of a cord.

BKpiH-5 elastic cord is mixture of BKHA (piH75 23B) powder, ruby powder, ruby CKTH and catalyzer no.18 which is drawn in form of cord.

ELECTRO PLATING SECTION

Electro platting is a process based on the principle of Faraday's law.

"Where material deposited (m) is directly proportional to product of the current (Q) and time (t)".

If time is less than the current supply must be more and if current is less than time required for deposition of material is more. This is an electrolysis process with anode (99.9% pure base metal) and cathode (component work piece).The supply is of DC current & not AC. If the alternating current is supplied then for one step it is ON position the deposition of base metal takes & other cycle removal of base metal takes place so net electro platting is zero in order to avoid this DC is supplied.

Requirements for electro platting process:

a) Electro platting set up

b) Power supply (DC)

c) Tank-water salt solution (generally AgCN, KCN, K2CO3) of metal to be deposited.

D) Anode-pure base metal

E) Cathode-work piece.

Process: When DC is supplied +ions from anode move to cathode and get deposited as a result on work piece there is a protective layer formation. Along with this H2 also get deposited on work piece later it is removed by other process.

Steps in electroplating:

A) Pretreatment-cleaning, dipping in a solvent, alkaline, mechanical (sand blasting).

b) Platting (electrolysis process)

c) Post treatment-De hydrogenation heating at certain temperature for certain time generally at 200 degree celsius for 2 hrs.

Some special process in electroplating shop are:

Electrophoresis-This process is to increase the frictional properties on dry surface of components like bush & lever. Components involved are lever, bush, end piece.

This a coating process is for depositing/impregnating graphite on the components of AL-31FP engine. The layer acts as a hard lubricant on stainless steels, nickel base alloys titanium alloys components. Dispersed graphite in water is used for the process. The thickness of layer is 6-9microns.The layer is fragile so that heat treatment is required after the process for strengthening.

Ni platting of Al powder (HA-67)-The powder is as a bond coat for AHb&YBC-2P plasma coating process. Components involved are rare casing, 1st, 2nd, 3rd stage casing & 1st stage IGV casing.

Ni platting of Al powder process, special coating process uses Al powdered-o to produce a powder HA-67

Impulse anodizing: Process is carried out to improve wear resistant of nut & brush. Equipment used is special platting bath of indigenous origin. Components involved are nut, couple nut, and union nut.

It is a coating process to impart hardness on the threaded portions of titanium parts of AL-31FP in electroplating shop. An anodic film is formed on parts by application of DC current under 60-200 volts.The current is regulated in impulse form with a frequency of 1-10Hz.Steal plate is used as cathode and mixture of H_2so_4 (200-210mL/L) & phosphoric acid (10-20mL/L) is used as electrolyte.

UBCP-3C Coating-This coating used as a high temperature lubrication coating, equipment used is drying oven & hair brush. Components used valve, casing piston, slotted tube.

It is a suspension consisting of heat resistant silicon organic varnish KO-815 cadmium oxide, graphite powder C-1 highly volatile. Solvent (xylene+acetone+ethyl cellosolve) this coating is applied using a brush or by standard sprayer. Heating is done in an oven by slowly heating component at rate of 1-3deg celsius/minute to 250degree celsius and held for two hours. The coating is applied nickel alloys, titanium alloys steel alloys.

BAP-2 coating-This coating is used as an ant frictional coating. Equipment used is drying oven hair brush. Components used are gear,sector,bush,pinion.

It is a suspension consisting of epoxide varnish EP-074, molybdenum disulphide powder highly volatile solvent (xylene+acetone+ethyl collusive).This coating is applied using a hairbrush. Heating is done in an oven by slowly heating the component at the rate of 1-3 degree celsius to 250degree celsius and held for two hours. The coating is applied on steel bronze based materials.

ASSEMBLY PROCESS

In **AL31FP** the assembly of different modules of the engine is done in stage by stage.

1) CBD alignment in the compressor intermediate casing with bearing supports.

2) Compressor intermediate casing is aligned with the High pressure compressor section on outlet side.

3) On inlet side Low pressure compressor module is aligned with inlet guide vanes.

4) Now outlet of HPC module combustion chamber section is aligned.

5) Now turbine support casing aligned separately and turbine is aligned to it on inlet side.

6) Now combustion chamber outlet is aligned to turbine inlet.

7) After, turbine outlet is aligned to the diffuser casing then afterburner with nozzle alignment is done.

In prior to these steps the casings and supports undergo for co-axially calculation in order to align the axis of different sections of stationary parts like outer casings, afterburner diffuser, intermediate casing ,turbine support.

The rotational parts like LPC, HPC, LPT, HPT, LP shaft, HP shaft, and flexible shaft undergo balancing in order to achieve the dynamic balancing of the modules and aligned.

A mass particle moving in circular path experiences a centripetal force and a force required to produce it .An equal and opposite force acting radially outwards acts on the axis of rotation and is known as centrifugal force. This is a disturbing force on the axis of rotation the magnitude of which is constant but the direction changes with the rotation of mass. In a revolving rotor the centrifugal force remains balanced as long as the center of the mass of rotor lies on the axis of shaft. when the center of mass does not lie on the axis or there is an eccentric unbalanced force is produced.

Effect-Force exerted on the frame by the moving machine members are time varying, imparts vibratory motion to the frame and produce noise.

Reduction-By redistribution of mass which is accomplished by the addition or removal of mass from various machine members.

Types of balancing:

Balancing is done in two types static and dynamic balancing

Static balancing-A system of rotating mass is said to be in static balance, if the combined mass center of the system lies on the axes of rotation. If the combined mass center exist only one side of axes of rotation, then static balancing takes place. If in any rotation system, centrifugal forces are balanced the system then the system is said to be statically balanced

Dynamic balancing: Several masses rotate in different plane, the centrifugal force in addition to being out of balance also form a couple balancing or dynamic balancing .A system of rotating masses is in dynamic balance when there does not exist any resultant centrifugal force as well as resultant couple.

$$m1r1=m2r2$$

Centrifugal force are balanced hence the system is dynamically balanced .But they form an unbalanced couple magnitude m1r1=m2r2

Due to this unbalanced couple the system becomes dynamically unbalances. Balancing at which it is determined and decrease the misbalance of rotor, charactering its dynamic misbalance.

Overhauling the engine

TTL-Total technical life for AL31FP it is 2000 hrs.

FOH -Full Overhaul which means the engine has completed its life time.
(For AL31FP the FOH is 1000 hrs.)
IOH – Interim Overhaul which means the engine experienced a defect before life time.

Any engine on completion/before completion its life time if it comes to workshop it is said to be overhauling engine .The extent of work is based on area and extent of damage happened to engine decided by methods section. Dismantling goes in step by step firstly the engine is dismantled as Opposite to assembly, after dismantling the engine parts are washed. The washing is three categories depending upon the size of parts and carbon deposited on them.

Category A: In this stage every engine part is soaked in Soda bath (soda used is PA-290

Versatile powder) for 3-4 hrs. At 80 degree centigrade with ultra-sonic vibrations so that
Carbon deposits are removed.

Category B: In this stage every part is flushed with Aviation Turbonic fuel (ATF) at 70 degree centigrade mainly in pipe sections at different pressures. This process is mainly for removing the blockages in parts.

Category C: In this stage every part is cleaned with petrol and naphtha, after drying parts they sent to the view room inspection for having report on the parts.

View room inspection inspects the physical condition and they make report to methods section with adopted technology whether the part can be overhauled or rejected. So on final clearance with method sections the parts are sent to respective shops for overhauling if they
Are not rejected after overhauling the parts. After satisfactory rework the parts are being accepted through quality control department and then sent back to assembly shop for engine buildup.

TEST HOUSE

Testing of the aero engine is the process of monitoring health & performance characteristics of engine by subjecting it to stimulations in flight conditions & there by tuning it to behave as per technological requirements. In the test house we have types of testing, parameters measured, test cell installation and test bed systems.

Types of AL31-FP engine testing: Aero engines are subjected to various test, which will guarantee reliable operation of engine during its service life. Here AL31FP engine is subjected to two tests and combustion chamber module of engine for temperature profile measurement test.

A) Initial test (check test), b) Final test (acceptance test), c) combustion chamber testing.

a) Initial test: This test is conducted for mainly to check the working of engine aggregates and some modules of the engines. Initial test as 3 stages

Stage 1-Check quality of assembly,

Running of engine, checking the technological actuation of Thrust vectoring control (TVC),

Vibrio graphing of engine BKA and KDA,

Checking of parameters of the engine & correcting to ISA conditions.

Stage 2-Fixing HP IGV as a function of n2 with T3 at n2=95%,

Fixing LP IGV as function of n1,

Final fixing of automatic actuation of jet nozzle,

Fixing frequency of rotation of n2 at idle ratting.

Stage 3-Checking and testing of turbine cooling aggregate,

Checking and stabilization of channels n1, n2&T4 of electronic engine controller aggregates,

Checking & adjustments of the centrifugal regulator of frequency of rotation n2 on unit NR-31BT1

Checking of jet nozzle operation at max & intermediate modes.

After Initial test if performance of engine is found satisfactory then the engine is sent to assembly shop for stripping &re assembly. The bearing of turbine blades are critically examined, minor defects are rectified, defective components are replaced and engine is repeated for initial test.

Final test: This test has 4 stages of which first 3 stages are same as initial test but after reassembly only.

Stage 4-Setting of pressure ratio across turbine and regulating of TVC

Checking & fixing of engine at variable modes by caring out acceleration/deceleration

Checking function of aggregate KD (anti surge)

Checking of operation of engine in reheat mode and emerging switching of reheat

Final fixing of centrifugal regulator and T4 channel of KPba

Checking re-arrangements of rotation devices of TVC at dry max reheat mode and transient mode

Combustion Chamber test: To check stability of manufacturing of the module, ensuring given radial profile and circular non-uniformity of temperature field of gas at outlet of combustion chamber.

C.C test is carried out prior to installation of the module on an initial test engine.

Internal preservation: All engines after initial & final test are subjected to internal preservation on test bed. During preservation a layer of preserving oil is formed inside the cavities of fuel aggregates & oil system, which helps forming corrosion formation & malfunctioning of aggregates. Fuel system aggregates are preserved with TN-321 oil and oil system aggregates of engine are preserved with TN210.Internal preservation is done when engine is on test bed. It is valid for a week without external preservation. If external preservation is done it has to be 48 hours from internal preservation and the preservation is valid for 5 years.

Parameters Measured: All the parameters which are observed corrected to ISA conditions. The parameters measured are

1) N1, N2 rpm at regimes idle, max & reheat.

2) T4 temperature after LP turbine at idle, max & reheat

3) Thrust and air mass flow in kg/s a regimes n2=88% max, reheat.

4) Specific fuel consumption in kg/hrs./kg

6) P2 static (compressor delivery pressure) in kg/cm2 to arrive at OPR for engine for regimes max & reheat.

In addition to above the starting acceleration characteristics are also are regulated & monitored at every stage of engine testing. If a defect which cannot be rectified in-site and which warrants withdrawal of engine is observed during the test, the engine is subjected to investigation, rectification & re-tests. The engine after inspection rectification is assembled and subjected to final test.

Test cell installation: All the aero engines are perfectly mounted with support in the test house. Engine is mounted on two beds which one is fixed to ground and the other fixed to first bed and it is movable, strain gauges are fixed to this and corresponding thrust can be calculated.

a) Control cabin-It consists of control panel, data acquit ion system (DAS).

b)Test cell-1)It has engine test bed which one is fixed and other flowable.2)Engine AAGB mounting system.3)System for discharge, exhaust gases.4)Protective device at engine air intake duct.5)Manifold pipe line of fuel & hydraulic lines for measuring fuel & oil pressure.6)Thrust measuring calibration system7)Compressed air system.8)Water cooling system for generator lubrication oil.9)Automatic fire extinguisher

c) Technological room-Hydraulic system preserver, Nitrogen in vacuum creation, Air cylinder control, NP-128 cooling system.

Test Bed Systems:

1) Fuel system: Fuel used is jet-A-1(ATF)

This system feed the fuel to engine, turbo starter at constant pressure irrespective of various operation, loading the booster pump (DCN-80) AAGB, feeding fuel to jet nozzle cylinders, IGV

2) Filtration system: 2 coarse filters to initial filtration, 3filters respectively of size 40mic, 5-8mic, 12-16mic to fine filtration

3) Lubrication system: Oil used is VINI IP, system caters oil to engine & GP-23/25(AC generator).

4) Hydraulic oil system (AMG-10): System serves for loading of hydraulic pump NP-128 to various flow & pressure requirements to stimulate actual load of aircraft system on pump.

Viscosity of oil <10cst, oil should be free from water, mechanical impurities.

5) Preservation oil system (TN-321 mineral oil): System serves for internal preservation of engine AAGB fuel/oil aggregates.

6) Water system: Serves for cooling of oil of hydraulic pump NP-128 & AC generator GP-23/25, for filtration of water it is gone through chemical treatment of reverse osmosis plant.

TESTING PARAMETERS

At test bed there are some testing parameters to be satisfied at the time of engine testing. For AL31FP the testing parameters is already stated in the ghost technology .If the engine on test bed satisfy these parametric values then the engine is perfect.

Initial test parameters

N1 %	**100%**
N2 %	**101%**
T3 k	**1645-1665**
T4 k	**<=1113**
P kg/cm2	**2.5-3.2**

Final test parameters

	N1 %	N2 %	Thrust Kgf	specific fuel consumption Kg/hrs./Kg	T 3 K	T 4 K	P Kg/cm^2	Air mass flow
Idle rating	Variable	**70**	**< 250**	**----------**	**---------**	**773**	**2.8-3.2**	**109.76-113.12**
Cruise rating	**-----------**	**85-90**	**3500-4400**	**<= 0.71**	**----------**	**----------**	**2.5-3.2**	**109.76-113.12**
Max rating (C.M)	**96-99**	**99-101**	**7478-7782**	**<=0.79**	**<=1665**	**<=1113**	**2.5-3.2**	**109.76-113.12**
Max rating (T.M)	**92.5-95.5**	**96-99**	**6929-7211**	**<=0.78**	**<=1590**	**<=1063**	**2.5-3.2**	**109.76-113.12**
Min reheat (C.M)	**96-99**	**99-101**	**<=8400**	**<=0.96**	**<=1665**	**<=1113**	**2.5-3.2**	**109.76-113.12**
Min reheat (T.M)	**92.5-95.5**	**96-99**	**<=7960**	**<=1.0**	**<=1590**	**<=1063**	**2.5-3.2**	**109.76-113.12**
Full reheat (C.M)	**96-99**	**99.2-100**	**12191-12440**	**<=1.97**	**<=1665**	**<=1113**	**2.5-3.2**	**109.76-113.12**
Full reheat (T.M)	**92-95**	**96.0-99.0**	**11123-11463**	**<=2.05**	**<=1590**	**<=1063**	**2.5-3.2**	**109.76-113.12**
Special mode	**100-103**	**101-103**	**12485-12740**	**<=1.95**	**<=1735**	**<=1168**	**2.5-3.2**	**109.76-113.12**

*C.M-combat mode. T.M-training mode.
* In idle rating fuel flow is checked to be 400-530 kg/hr.

Comparative study table of HAL koraput manufactured aero engines

Parameters		R11F20 /2s/2sk	R13-300	R25	R29B	RD33	AL31 FP	AL55I
Compressor stages LP&HP		3+3 axial	3+5 axial	3+5 axial	5+6 axial	4+9 axial	4+9 axial	3+5 axial
Turbine LP&HP		1+1	1+1	1+1	1+1	1+1	1+1	1+1
Comps ratio		8.9	8.75-8.9	9.2-9.5	12.2	20-22	3.5*6.6	17.55
Air mass flow kg/s		66.2	66-67	68.5	105	76+1.5	112	29.16
N1 (100%)		11,150	11,156	11,156	8,482	11,000	10,098	16,200
N2 (100%)		11,412	11,362	11,362	8,772	15,511	13,300	24,450
Combustion chamber		Can annular	Can annular	Can annular	Annular	Annular	Annular	Annular
(T3)		936	970	1040	1130	1257	1392	1172
T4		730	780	850	840	800	765	899
Thrust(Kgf)	Dry	3900	4100	4100	8000	5040	7670	1760
	Afterburner	6175	6600	7100	11500	8300	12500	--------
S.F.C(kg/hrs./kg)	Dry	0.99	.96	.96	.94	.77	.67	.69
	Afterburner	2.37	2.25	2.25	1.8	2.05	1.96	--------
S.Thrust(kgf/kg/sec)	Dry	59.81	62.12	60.29	76.19	66.3	68.48	--------
	Afterburner	94.70	100.98	110.73	109.5	107.7	111.61	--------
Vibration (mm/sec) or g	Compressor	3.5g	2.5g	2.5g	2.5g	30-40 mm/sec	50-70 mm/sec	40 mm/sec
	Turbine	4.5g	2.5g	2.5g	3.0g	30-40 mm/sec	50-70 mm/sec	40 mm/s
Length (mm)		4600	4600	4615	4991.5	4106	4945	1950

Max.dia(mm)		906	907	907	986	1000	1220	462
Weight(kg)	Dry	1128.5	1167+2%	1210+2%	1777+2%	1055	1530	385+5
	Dispatch	1205.25	1245+2%	1289+2%	1943+2%	1217.4	1530	385+5
Thrust to weight ratio	Dry	3.463	3.388	3.388	4.5	4.13	5.013	-------
	Afterburner	5.483	5.508	5.508	6.5	6.82	8.169	--------
Bypass ratio	--------	---------	--------	-------	------	.45	.571	.55
TBO/TTL (hrs.)		250/1650	500/1500	400/1200	550/ 1500	300/ 1000	1000/ 2000	1200 / Cold - 6000 Hot- 3600
Article		37	95	25	55B	88	96	
Aircraft		MIG 21FL/M/MF	MIG21M/MF	MIG21 BIS	MIG 27 M/23BM	MIG 29	SU 30 MKI	IJT-HJT 36
Indigenous name		Badal	------	Vikram	Rakhshak/Bahadur	Baaz/Garud	------	NPO saturn
Design bureau		Tumansky	------	Motor	Tumansky	Klimov	Lulka	NPO saturn lulka
Generation		2	2	2	3	4	4.5	------

CENTRAL LABARATORY

Central laboratory is testing the strength of parts, impurities present in the parts and gives the final report whether the parts be further used or rejected.

It has five sections-

1) Metallographic section
2) Mechanical testing shop
3) Pyrometer
4) Fuel and oil section
5) Chemical section (organic & inorganic)

1) **Metallographic section**: This section gives the micro level mechanical properties, grains orientation by finding crystal orientation angle using x-ray diffraction method

2) **Mechanical testing shop**: This section includes tensile test, compressive test, fatigue test, creep test and impact test

Tensile/compressive test-Here the specimen of different parts are taken and tensile/compressive load is applied and find its tensile/compressive strength

Creep test-This test is used to test the components which work in high temperature. Here the component is heated in closed furnace for some time till the required temperature is attained and same tensile/compressive test is applied and find tensile/compressive strength at high temperature.

Fatigue-When the blades move with the high rpm then due to centrifugal force there may be chances of blades to elongate and may touch the casing and damage the blades, so this test gives the range of elongation of blades for the respective time & frequency. In this test first the blade is set in to the resonance by certain frequency and find the elongation range by using microscope. So according to this range the blade can be rotated in fixed range and avoiding damage of blade & engine.

Impact test-Here the specimen of component is fixed in the apparatus and sudden load is applied and find the impact load which the component can sustain.

3) **Pyrometery: This** section uses science of measuring temperature.

1) Contact method-

a) By expansion thermometer (solid expansion bi-metallic strip, liquid expansion by mercury in glass thermometer)

B) Thermo electric pyrometer thermocouples

c) Electrical resistance thermocouples

d) Changing rate of testing body by crayons & temperature.

2) Non-contact method-

a) By total radiation pyrometer

B) Optical pyrometer

c) Photo electric pyrometer

4)**Fuel and oil section**: In this section the class(purity) of ATF is measured by different process like flash point, specific gravity of fuel, purity of oil checking(lubricating oil , hydraulic oil),acidity checking, viscosity of oil with varied temperature , alkalinity ,drop point of wax and grease, penetration test.

SPECTROSCOPY ANALYSIS

Spectroscopy analysis is carried out to check the metallic composition of given metal basing on the principle of emission and absorption spectra.

The apparatus is designed in a way that it can detect 30 elements which are already placed in grooves of the apparatus. When the detective metal is placed in observation box a spark is generated in and electrons are emitted from the surface of metal. The emitted electrons absorbs some amount of energy to excite to higher energy levels. As the electrons are not of stabilized they immediately de excite to lower energy levels and energy is released in the form of light. There is the formation of spectra and the metallic absorbers placed in grooves absorb the light and indicates the metallic composition. The percentage of composition is based on the intensity of band formed in the spectra.

X-RAY DIFFRACTION

X-ray diffraction is used to know the crystallographic orientation of crystals present in the metal on the basis of Brags law of diffraction.

Wavelength=2*D*sin (angle)

The wavelength is x ray wavelength and distance from the slit is known and the angle is calculated.

This specially used in finding the orientation angle of grain seeds in single crystal investment casting.

<u>Nondestructive test (NDT)</u>-

Luma and dye-This test is used to detect the

1) Process flaws

2) Service flaws

3) Inherent flaws (casting, forging)

4) Surface, sub surface flaws.

Luma check-First the part is cleaned with petrol, acetone for 5 minutes then part is preheated in oil (T=100 degree Celsius) which avoids moisture formation on the part. After preheating it is allowed to cool for 15 minutes. Then the parts are dipped in Zyglo (liquid form) for 10 minutes, zyglo moves in to flaws and remain there and after dipping in zyglo part is washed and set to dry, developer (powder form) is sprayed on the parts and moved to ultraviolet test, here the developer glows when UV light is passed by which the flaws present are detected.

Red Dye check-Here the process is same as the luma check but instead of part dipping in zyglo it is dipped in the red dye (SKL-WP) and developer is sprayed and moved to UV test where the flaws are detected.

Magnetic field test-In the luma and dye check only surface, subsurface flaws are detected but the flaws in inner surface ,circular parts are detected by magnetic field test. It works on the principle of right hand thumb rule, if current moves in straight direction on the surface of part then magnetic field moves circular direction to the part and vice versa. Magnetic field gets deflected and creates the south and north poles if it detects any flaw in its direction. For circular parts current is supplied in straight direction and for straight parts current is in circular direction to its surface.

Eddy current test-This test is used to find cracks only on the edges of blades and not the profile of the blades. Here blades like1 to 9th HPC stators, inlet guide vanes, LPC stators & rotors from 1 to 4 stages and all turbine blades are checked. The pencil probe is moved on the edge of the blade and sounds for red light(long beep) it indicates the crack on edge of blade, if not it sounds for green light(single beep) no crack is present, the beep sound can be observed with the help of headset. There are different types of probes used for this purpose.

Ultrasonic test-This test is used to find internal flaws, it uses ultrasonic waves and defects can be found by sound waves reflection, it uses probe of piezoelectric piston connected to instrument which gives directly the readings in the form of graph .First the liquid is applied on the surface of the part as liquid is good carrier of sound waves. Ultrasonic test is used for checking the discs, shafts of LP & HP.

X-ray radiography-This test works on the principle of X-ray analysis in medicine with a voltage variation around 200 volt due to have high intensity x-ray. It is used to study subsurface defects of the components.

CONCLUSION

This internship project helped us a lot to understand about various components of jet engines, and their assembly, manufacturing and testing of engine.

We have got handful of virtual experience over the components,their purpose, functioning and design.We have observed how the generations of combat aircraft engines developed and undergoing development.

The design of this 4+ generation jet engine is mainly concentrated over the intermediate casing in order to improve by pass ratio which enhance the performance of engine as well as engine life span when compared to other engines its life span s enhanced by 40%.And also the TVC system with variable jet nozzle as to guide the exhuast gases in order to make the aircraft superior in combat by its marvelous manouvering capability.

The assembly of jet engine is complicated,AL31FP is more complicated due its design and aggregates mounted on it with over 60,000 parts.In assembly section two more bays plays a vital role.One is balancing bay where it balances the rotating parts of engine and other is co-axiality bay where it maintains the coaxiality of stationary parts of engine.These two bays always maintain there work precision approimately towards zero error in order to avoid the damage of engine due to vibrational effect produced during flight.

Manufacturing of jet engine componets is high precised process which makes the production process more costly due to usage of high precision CNC machines in production process.

Testing of engine ensures for assembly of engine in aircraft for flight.So it is also carried with zero tolerance of various parameters during the test.In testing various components and aggregates functioning is checked and the adjustments are made to eradicate the malfunction of engine during flight.

Not only these sections there are many sections in order to produce the jet engine with high precision.Some sections like in this division is central laboratary in order to analyze the chemical and metaloghrapic analysis for the alloys.R&D wing plays vital role for providing solution in practical problems and implementing the design of the engine.

FUTURE SCOPES

There is a wide range of scope in R&D section of jet engines in order to support the modern aviation.As we go for the combat aircraft generation India is having 4+ generation i.e SUKHOI-30MKI with AL31FP engine mainly concentrated over the bypass ratio and TVC system with 14 degree tilting in up and down with alignment of 32 degree to engine axis.

At present scientists are working for 5 generation combat aircraft engine which is mainly concentrated over the TVC with tilting in 180 degree with the 2 axis of engine.

Major scope lies in the metallurgy section in order to choose the best alloy which meets the requirement and enhances the engine life and pays contribution to reduce the engine weight.

Next it lies in the production technology because most of the cost adds in production department when it is compared with raw material of engine.

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